

Naive Set Theory

DoremySwee

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AI-Usage: THIS ARTICLE IS AI-TRANSLATED FROM CHINESE VERSION.

There is a quick manual check, but not that accurate due to lack of time - this text is originally written to help Chinese students poor at English to get started on undergraduate Maths.

This note is designed for those with basic logic abilities, and starting their undergraduate math.

1 Functions

We will start with the most fundamental question. What is a function? High school textbooks tell us that a function is a correspondence between input and output. Given a set X called the domain, we say f is a function with domain X if and only if for each $x \in X$, $f(x)$ is always a uniquely determined mathematical object.

But this definition is very... inelegant. It only tells us what a function is, doesn't tell us how to ensure a function exists, and requires an additional definition to tell us when two functions are the same function. Unlike concepts like "triangle" or "quadratic equation in one variable" which are very clear – you can draw a triangle as long as you have three vertices, and you can write a quadratic equation as long as you have three coefficients. We hope such vague things are as few as possible in mathematics. Mathematicians ultimately chose to represent everything using sets, leaving only "set" as an abstract mathematical object that requires a large number of axioms to characterize.

When does a function exist... Let me ask you, do indescribable functions exist? Does there exist a discontinuous function that inputs a real number and outputs a real number, such that $f(x + y) = f(x) + f(y)$? We can't even write down its defining expression, so does it actually exist?

Let's leave the question of when functions exist to the set theorists:

Definition 1.1 (Set theoretic function). *A set theoretic function is a set f consisting of ordered pairs, such that for any mathematical object x , there exists **at most** one mathematical object y such that $(x, y) \in f$. In this case, we can denote y by $f(x)$. In other words, a set theoretic function can be viewed as $\{(x, f(x)) \mid x \in \text{domain}\}$. The domain of a set theoretic function is the set of all x satisfying $\exists y, (x, y) \in f$.*

Two set theoretic functions f, g are equal if and only if they are equal as sets, or equivalently, if they have the same domain X , and $\forall x \in X, f(x) = g(x)$.

In axiomatic set theory, we also try to define ordered pairs as sets: $(x, y) = \{\{x\}, \{x, y\}\}$ is a common definition. In particular, for (x, x) we can simplify it, writing it as $\{\{x\}\}$.

Definition 1.2 (Codomain). *We often want to know what $f(x)$ is – is it a real number, a vector, or some other strange object? So we generally want a set Y , ensuring $\forall x \in X, f(x) \in Y$. In this case, we write: $f : X \rightarrow Y$, meaning f inputs an element from X and outputs an element from Y .*

To be more rigorous, we define a function as an ordered pair (f, Y) , where f is a set theoretic function, X is its domain, and it satisfies $\forall x \in X, f(x) \in Y$.

It is worth noting that the codomain is not necessarily the range. For example, $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ has a range of non-negative real numbers, while $f : \mathbb{R} \rightarrow \mathbb{R}$ already specifies its codomain as $\mathbb{R}$.

Also worth noting is that two functions are equal if and only if their codomains are equal and their corresponding set theoretic functions are equal.

The aboves are not rigid rules, but are actually for the convenience of our daily use. When defining functions, we hope to specify what $f(x)$ is, but we might not have a deep enough understanding of the function to know what the range is; on the other hand, we also want to discuss the relationship between the domain and codomain using functions, so the codomain is important. In the next section, we will use functions to discuss the size of sets.

Definition 1.3 (Sequence). *A sequence in a set S is a function with domain the set of natural numbers or positive integers, and codomain S – e.g., a real sequence requires $S = \mathbb{R}$.*

2 Size of Sets

2.1 Definition and Basic Corollaries

Definition 2.1 (Injection, Surjection, Bijection). *A function $f : X \rightarrow Y$:*

- is an injection if and only if different x correspond to different $f(x)$:

$$\forall x_1, x_2 \in X, (x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2))$$

If we transform this using the contrapositive, we get another definition:

$$\forall x_1, x_2 \in X, (f(x_1) = f(x_2) \Rightarrow x_1 = x_2)$$

- is a surjection if and only if every $y \in Y$ has a corresponding x :

$$\forall y \in Y, \exists x \in X, f(x) = y$$

- is a bijection if and only if it is both an injection and a surjection.

Note that the definition of surjection is based on the codomain. A function is a surjection if and only if its codomain is equal to its range.

Let's first elaborate on the relationship between injections and surjections:

Proposition 2.1. If $f : X \rightarrow Y$ is an injection, and $X \neq \emptyset$, then for any $x \in X$ we can construct a surjection $g_x : Y \rightarrow X$:

$$g_x(y) = \begin{cases} x, & f(t) = y \\ t, & y \notin f\text{'s range} \end{cases}$$

Conversely, if $g : Y \rightarrow X$ is a surjection, we just need to choose. For each $x \in X$, choose one $y \in Y$ such that $g(y) = x$, and then define $f(x) = y$ to get an injection $f : X \rightarrow Y$. If X is a finite set, we can complete this selection in finitely many steps, constructing the set theoretic function corresponding to f . But if X is an infinite set, then it's troublesome – we must use the famous Axiom of Choice to construct the set theoretic function corresponding to f .

However, the Axiom of Choice is a cornerstone of modern mathematics, so feel free to use it. We will discuss it separately later.

Definition 2.2. We say a set X is smaller than a set Y if and only if there does not exist an injection $f : Y \rightarrow X$. In this case, we say Y is larger than X . If X is not larger than Y , and not smaller than Y , then we say X and Y are the same size.

Exercise 2.1. *For finite sets, verify that the above definition conforms to convention.*

Exercise 2.2. *Based on the previously explained relationship between injections and surjections, we can change the definition above from "no injection" to "no surjection in the opposite direction" – though we need to handle the empty set separately. Please rewrite the definition accordingly.*

Theorem 2.1 (Cantor–Bernstein–Schroeder Theorem). *Two sets are the same size if and only if there exists a bijection between them.*

Proof. The proof is relatively complex and will not be detailed here. However, the proof does not involve any advanced definitions, theorems, or techniques; it only requires some logical proficiency. If interested, you can try to prove it yourself and search online to check your proof. $\square$

2.2 Countable Sets

We have established a basic understanding of set size (though there are still quite a few omissions...). Let's first look at some basic examples:

Example 2.2. *The set of positive integers $\mathbb{Z}^+$ and the set of positive even numbers $2\mathbb{Z}^+$ are the same size.*

Proof. The function $f : \mathbb{Z}^+ \rightarrow 2\mathbb{Z}^+, f(x) := 2x$ is a bijection. $\square$

Example 2.3. *The set of natural numbers and the set of integers are the same size.*

Proof. The function $f : \mathbb{N} \rightarrow \mathbb{Z}, f(x) := \begin{cases} \frac{x}{2} & x \text{ is even} \\ -\frac{x+1}{2} & x \text{ is odd} \end{cases}$ is a bijection. $\square$

Example 2.4. *The set of natural numbers and the set of rational numbers are the same size.*

Proof. Our immediate task is to split a natural number into two. There are many methods; you can search online. I'll use a more number-theoretic method here: for example, for any natural number n , there always exists a unique pair of natural numbers (m, k) such

that $n + 1 = 2^k(2m + 1)$. In other words, this relation establishes a bijection between the set of natural numbers and $\{(m, k) \mid m, k \in \mathbb{N}\}$.

Next, we can use this relation to establish a surjection from the natural numbers to the rational numbers: first, using the bijection from the previous example, transform m into an integer t , then define:

$$g(n) := \begin{cases} \frac{t}{k}, & k \neq 0 \\ 0, & k = 0 \end{cases}$$

Clearly, $g : \mathbb{N} \rightarrow \mathbb{Q}$ is a surjection. Therefore, $\mathbb{N}$ is not smaller than $\mathbb{Q}$. At the same time, there exists an injection $\iota : \mathbb{N} \rightarrow \mathbb{Q}$, $\iota(n) := n$, so $\mathbb{N}$ is not larger than $\mathbb{Q}$. Therefore, $\mathbb{N}$ and $\mathbb{Q}$ are the same size. $\square$

Remark 2.5. *Sometimes you may need to split a natural number into any finite number of parts. The method is to directly factorize $n + 1$ to get $n + 1 = 2^{m_1}3^{m_2}5^{m_3} \dots$, obtaining any finite number of natural numbers, as in the following exercise:*

Exercise 2.3. *We say a function $f : \mathbb{N} \rightarrow \mathbb{N}$ is decreasing if and only if $\forall n \in \mathbb{N}, f(n) \geq f(n + 1)$. All decreasing functions form a set S . Prove that S is the same size as $\mathbb{N}$.*

Definition 2.3 (Countable Set). *A set that is the same size as the set of natural numbers is called a countable set. Depending on the context, some authors may also include finite sets when they say countable set. An uncountable set definitely refers to a set larger than the set of natural numbers.*

2.3 Cantor's Diagonal Argument

Theorem 2.2. *The set of real numbers is larger than the set of natural numbers.*

Proof. Search online. In fact, a common proof has a minor flaw related to $0.\dot{9} = 1\dots$ $\square$

Since I don't want to discuss that trivial flaw, nor leave a flaw in the lecture notes, let's directly see how to construct a larger set for any given set:

Definition 2.4 (Power Set). *For a set X , we consider the set of all its subsets, called $\mathcal{P}(X)$.*

Theorem 2.3. $\mathcal{P}(X)$ is always larger than X .

Proof. Assume there exists a surjection $g : X \rightarrow \mathcal{P}(X)$. Define a subset S of X : $S := \{x \mid x \notin g(x)\}$. Since g is a surjection, assume $g(s) = S$. Is s an element of S ? If yes, then by definition $s \notin g(s)$, contradiction; conversely, if $s \notin S$, then by definition $s \in g(s)$, also a contradiction. So the assumption that g is a surjection must be wrong. Therefore, such a surjection certainly does not exist. $\square$

3 Basic Theory of Functions

3.1 Composition of Functions

We often need to discuss functions abstractly, discussing several functions $f, g, h, \dots$ without discussing $f(x)$. In such cases, we need the "composition of functions" to combine functions. If you have studied matrices, I can tell you that matrices are essentially functions, and matrix multiplication is the function composition we are about to introduce.

Definition 3.1 (Composition of Functions). *For functions $f : X \rightarrow Y, g : Y \rightarrow Z$, we define the function $g \circ f : X \rightarrow Z$:*

$$(g \circ f)(x) := g(f(x))$$

You might think that discussing f without discussing $f(x)$ is far from your experience, but that's wrong. Look at the statement of the following theorem:

Proposition 3.1. *If set X is not larger than Y , and Y is not larger than Z , then X is not larger than Z . In other words, the composition of surjections is a surjection, and the composition of injections is an injection.*

However, the proof details still require using $f(x)$... but the statement itself, not involving $f(x)$, achieves a level of abstraction and simplification. Remember, abstraction is actually for simplification.

Exercise 3.1. *Prove the above proposition.*

And similar to the identity matrix, for convenience, we define the following function for each set:

Definition 3.2 (Identity Map). *For a set X , there exists a unique function $f : X \rightarrow X$ such that $\forall x \in X, f(x) = x$. Denote this function as id_X . It is clearly a bijection. Obviously, $\forall X, Y, \forall g : X \rightarrow Y, g \circ \text{id}_X = g = \text{id}_Y \circ g$.*

Definition 3.3 (Inclusion Map). *If $A \subseteq B$, then there exists a unique function $\iota : A \rightarrow B$ such that $\forall a \in A, \iota(a) = a$. This function is called the Inclusion Map, generally denoted by the letter ι .*

Definition 3.4 (Restriction of a map). *For a function $f : X \rightarrow Y$ and a subset $A \subseteq X$, there exists a unique function $g : A \rightarrow Y$ such that $\forall a \in A, g(a) = f(a)$. Denote this function as $f|_A$, called the restriction of f to A .*

Function restriction is also a very commonly used concept. However... we can actually define function restriction using the inclusion map. It's a kind of brain teaser, just play around with it:

Theorem 3.1. *For a function $f : X \rightarrow Y$ and a subset $A \subseteq X$, there is an inclusion map $\iota : A \rightarrow X$. We have the following equation:*

$$f|_A = f \circ \iota$$

3.2 Pre-Image

In this section, we define the pre-image of a function. This is both a common mathematical tool and **an excellent opportunity for you to practice transforming naive set problems into basic logical problems. Please study every proof carefully and complete the exercises.**

Definition 3.5 (Pre-Image). *Given a function $f : X \rightarrow Y$ and a subset $A \subseteq Y$, we define the pre-image:*

$$f^{-1}(A) := \{x \in X \mid f(x) \in A\}$$

This notation conflicts with the "inverse of a bijection". However, in everyday mathematics, you will certainly be able to distinguish between a subset of Y and an element of Y , so it's not a big problem as long as it's understood correctly.

Proposition 3.2. *For a function $f : X \rightarrow Y$, $f^{-1}(A)$ is a subset of X , satisfying the following properties:*

- $f^{-1}(\emptyset) = \emptyset, f^{-1}(Y) = X$
- $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$ (complement $Y \setminus A := \{y \in Y \mid y \notin A\}$)
- $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$
- $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$

Proof. • First, $f^{-1}(A) \subseteq X$.

- Obviously, $\forall x \in X, f(x) \notin \emptyset$.
- So $\forall x \in X, x \notin f^{-1}(\emptyset)$.
- Summarizing, $f^{-1}(\emptyset) \cap X = \emptyset$.
- Since $f^{-1}(\emptyset) \subseteq X$
- Therefore $f^{-1}(\emptyset) = \emptyset$

Similarly,

- Obviously, $\forall x \in X, f(x) \in Y$.
- So $\forall x \in X, x \in f^{-1}(Y)$.
- Summarizing, $f^{-1}(Y) \cap X = X$.
- Since $f^{-1}(Y) \subseteq X$
- Therefore $f^{-1}(Y) = X$
- – We want to prove $\forall x \in X, x \in LHS \Leftrightarrow x \in RHS$.
- First, $x \in LHS \Leftrightarrow f(x) \in Y \setminus A$
- Because $f(x)$ is always an element of Y , we have $f(x) \in Y \setminus A \Leftrightarrow f(x) \notin A$
- And $f(x) \notin A \Leftrightarrow x \notin f^{-1}(A)$
- Also, since $x \in X$, we have $x \notin f^{-1}(A) \Leftrightarrow x \in X \setminus f^{-1}(A) = RHS$.
- In summary, $\forall x \in X, x \in LHS \Leftrightarrow x \in RHS$, meaning $LHS \cap X = RHS \cap X$. And since LHS, RHS are both subsets of X , we have $LHS = RHS$.
- – We want to prove $\forall x \in X, x \in LHS \Leftrightarrow x \in RHS$. After proving this, we can use the final "X subset" argument as before to complete the proof of the entire proposition.

– For $x \in X$,

$$x \in LHS \Leftrightarrow f(x) \in A \cap B \quad (1)$$

$$\Leftrightarrow f(x) \in A \text{ and } f(x) \in B \quad (2)$$

$$\Leftrightarrow x \in f^{-1}(A) \text{ and } x \in f^{-1}(B) \quad (3)$$

$$\Leftrightarrow x \in RHS \quad (4)$$

- The proof for $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$ is left as an exercise.

□

In fact, the above intersection and union theorems also hold for infinite intersections and unions – unfortunately we haven’t covered that yet, otherwise you would have seen that version.

Exercise 3.2. *Prove: $(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A))$.*

3.3 Inverse Functions

Definition 3.6 (Inverse of a bijection). *Given a bijection $f : X \rightarrow Y$, there exists an inverse function $g : Y \rightarrow X$, denoted f^{-1} , such that $f \circ f^{-1} = id_Y$ and $f^{-1} \circ f = id_X$.*

Theorem 3.2 (Inverse Functions and Bijections). *$f : X \rightarrow Y$ is a bijection if and only if there exists a function $g : Y \rightarrow X$ such that g can serve as the inverse of f , satisfying: $f \circ g = id_Y$ and $g \circ f = id_X$*

Exercise 3.3. *Prove the above conclusion.*

Sometimes, rather than proving a function is both surjective and injective, constructing the inverse function and proving it is the inverse is simpler. And actually, the relationship of ”inverse function” is, in most cases, especially in algebra, more meaningful than ”both surjective and injective”.

4 Basic Set Theory

4.1 Collections of Sets

Definition 4.1 (Collection). *A collection generally refers to a set whose elements are all sets. This concept is not essential in mathematics but is proposed for the convenience of communication.*

For example, when discussing a topology problem, we consider points in a space forming some point sets, and we may need to handle many point sets simultaneously. The set formed by these point sets is called a collection, to distinguish it. In daily life, we rarely encounter sets that are "higher level" than collections, unless you are engaged in set theory related research.

Example 4.1. In axiomatic set theory, all mathematical objects are defined by pure sets. Clearly, the empty set is a pure set, then we have these sets:

$$\{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\{\emptyset\}\}, \dots$$

Mathematicians also designated some sets as "natural numbers": $0 := \emptyset, n+1 := n \cup \{n\}$, for example $3 = \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$.

Definition 4.2 (Indexed Collection). An indexed collection is similar to a collection, but it has an index. An indexed collection refers to a set theoretic function X with domain the index set I , and for each $i \in I$, this set theoretic function specifies a set X_i .

We will use collections and indexed collections immediately in infinite intersections and unions.

4.2 Infinite Intersections and Unions

Definition 4.3 (Union of a (Indexed) Collection). For a non-empty (indexed) collection, we define its union:

$$\bigcup \mathcal{C}^{collection} := \{x \mid \exists X \in \mathcal{C}, x \in X\}$$

$$\bigcup_{i \in I} X_i := \{x \mid \exists i \in I, x \in X_i\}$$

Such a union exists uniquely, its existence guaranteed by axioms.

Exercise 4.1. Verify that this definition is consistent with binary union.

Example 4.2. In this example, (a, b) denotes an open interval of real numbers.

- $\bigcup_{n \in \mathbb{N}} (0, n) = \mathbb{R}^+$
- $\bigcup_{n \in \mathbb{N}} [-1, -1/n] = [-1, 0)$

Exercise 4.2. Carefully verify the above examples, especially the boundaries of the intervals.

Definition 4.4 (Intersection of a (Indexed) Collection). *The intersection of an empty collection... cannot really be defined. We only define the intersection for non-empty collections.*

$$\mathbb{M}\mathcal{C}^{collection} := \{x \mid \forall X \in \mathcal{C}, x \in X\}$$

$$\bigcap_{i \in I} X_i := \{x \mid \forall i \in I, x \in X_i\}$$

Example 4.3. In this example, (a, b) denotes an open interval of real numbers.

- $\bigcap_{x \in \mathbb{R}^+} [0, x] = \{0\}$
- $\bigcap_{n \in \mathbb{N}} (n, \infty) = \emptyset$

Definition 4.5 (Disjoint Union). *For an indexed collection, we sometimes want to obtain a "union" where the sets in the collection are in some sense treated as mutually disjoint subsets of this union.*

$$\bigsqcup_{i \in I} X_i := \bigcup_{i \in I} \{(i, x) \mid x \in X_i\}$$

And define the "inclusion map" (clearly an injection):

$$\iota_i : X_i \rightarrow \bigsqcup_{i \in I} X_i, \iota_i(x) := (i, x)$$

If we treat X_i as the range of ι_i , then we successfully treat X_i as a subset of $\bigsqcup \dots$, and they are mutually disjoint.

That's right, this way if $X_1 := S, X_2 := S$, then $\bigsqcup_{i \in \{1, 2\}} X_i$ contains two copies of S rather than one.

Remark 4.4. Our disjoint union here is to avoid the original sets intersecting each other, so we "copy" each of them. Sometimes, disjoint union represents the direct union, but it implicitly indicates that the original sets are disjoint – for example, some authors write $S := A \sqcup B$ to mean $S = A \cup B$ and $A \cap B = \emptyset$.

Exercise 4.3. Rewrite the formulas concerning intersections and unions in the properties of pre-images into versions for infinite intersections and unions, and prove them.

4.3 n-Dimensional Space and Cartesian Product

Definition 4.6 (Direct Product / Cartesian Product). We have seen $\{(a, b) \mid a \in A, b \in B\}$ many times, let's define it specifically:

$$A \times B := \{(a, b) \mid a \in A, b \in B\}$$

Called the Cartesian product of A and B .

Remark 4.5. If a function has multiple inputs, e.g., $f(x, y) = x * y$ where $x \in \mathbb{R}, y \in \mathbb{R}$, we generally consider the domain of f to be the Cartesian product of the domains of each input variable $\mathbb{R} \times \mathbb{R}$. Also, binary operations like addition and multiplication can be viewed as functions with two input variables, or as functions $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$.

Example 4.6. That's right, the name Cartesian product comes directly from the Cartesian coordinate system. The simplest way to define three-dimensional space mathematically is to turn things upside down, using coordinates, which are originally used to describe points in space, to replace the points, defining space as the set of coordinates. The plane is $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$, three-dimensional space is $\mathbb{R}^3 := \mathbb{R}^2 \times \mathbb{R}$, and so on we can define n -dimensional space $\mathbb{R}^n$.

Remark 4.7. It seems the Cartesian product is not commutative... $(a, (b, c)) \neq ((a, b), c)$. But no one cares... Mathematically, there are methods to handle such negligible differences; we "redefine equality", and then they can become "completely identical" mathematical objects participating in other discussions.

Definition 4.7 (Infinite Cartesian Product). We can generalize the binary Cartesian product to get finite Cartesian products, but it neither has associativity nor can it be generalized to infinity. Given an indexed collection $(X_i)_{i \in I}$, define its Cartesian product:

$$\prod_{i \in I} X_i := \left\{ f : I \rightarrow \bigcup_{i \in I} X_i \mid \forall i \in I, f(i) \in X_i \right\}$$

Also define the projection function $\pi_i : \prod_{i \in I} X_i \rightarrow X_i$ (clearly a surjection):

$$\pi_i(f) := f(i)$$

Example 4.8. We sometimes want to consider the set of all functions from X to Y . Actually, just define a collection $(X_y)_{y \in Y}, \forall y \in$

$Y, X_y := X$, then its Cartesian product is the set of all functions from X to Y , denoted:

$$Y^X$$

Well, it is a bit convoluted. But this notation is quite reasonable – assuming the finite set Y has n elements and X has m elements, then Y^X indeed has n^m elements (in set theory, $0^0 = 1$).

5 Equivalence Relations and the Axiom of Choice

5.1 Definition and Basic Examples

We mentioned earlier that we want to force some different things to be considered the same, to simplify further discussions.

Example 5.1. Consider the set of integers $\mathbb{Z}$ and a positive integer n . We define $a \equiv a'$ if and only if $a - a'$ is an integer multiple of n . In this case, we say a and a' are equivalent.

We want to continue discussing addition and multiplication based on this equivalence (i.e., modular arithmetic), so we need to verify the following properties:

- For equivalent $a \equiv a', b \equiv b'$, is $a + a' \equiv b + b'$?
- For equivalent $a \equiv a', b \equiv b'$, is $aa' \equiv bb'$?

Fortunately, both properties are not difficult to prove with middle school mathematics. However, the annoying thing is that $<$ is not compatible with congruence $\equiv$:

$$\text{For } a \equiv a', b \equiv b', a < b \text{ does not imply } a' < b'$$

We need to generalize this concept of "equivalence" and provide a theory to verify whether the original structure is compatible with the equivalence structure.

Definition 5.1 (n -ary relation). For a set X , an n -ary relation can be viewed as a subset $R \subseteq X^n$. If $(x_1, x_2, \dots, x_n) \in R$, we write $R(x_1, x_2, \dots, x_n)$.

Equivalence relations and order relations are special binary relations, and are generally written in the form aRb , where R is replaced by a symbol.

Definition 5.2 (Equivalence Relation). *A binary relation $\equiv$ is an equivalence relation if and only if the following three properties hold:*

- *Reflexivity:* $\forall x \in X, x \equiv x$
- *Symmetry:* $\forall x, y \in X, (x \equiv y \Leftrightarrow y \equiv x)$
- *Transitivity:* $\forall x, y, z \in X, (x \equiv y \text{ and } y \equiv z) \Rightarrow (x \equiv z)$

Example 5.2. • *For a set X , we can define two rather meaningless equivalence relations – one says $x \equiv y$ if and only if $x = y$, the other says $x \equiv y$ always holds.*

- *It is not difficult to verify that the congruence relation on the integers is an equivalence relation.*
- *We can define an equivalence relation on the real numbers: $a \equiv b$ if and only if $a - b \in \mathbb{Q}$.*
- *Suppose there is a collection $\mathcal{C}$ where the sets are pairwise disjoint and $\emptyset \notin \mathcal{C}$. Such a collection is called a **partition**. We can define an equivalence relation on the set $\mathbb{U}\mathcal{C}$: $a \equiv b$ if and only if $\exists X \in \mathcal{C}, a \in X \text{ and } b \in X$.*

Exercise 5.1. *Carefully verify the last two examples.*

Definition 5.3 (Equivalence Class, Quotient Set). *After defining an equivalence relation, we want to squeeze all equivalent elements together into one... and the new set obtained is called the quotient set, where we discuss problems. To do this, first we define the equivalence class. For a set X and an equivalence relation $\equiv$ on it, the equivalence class of an element x is defined as:*

$$cl_{\equiv}(x) := \{t \in X \mid t \equiv x\}$$

Clearly, the equivalence classes of two elements are equal if and only if they are equivalent; two elements are not equivalent if and only if their equivalence classes are disjoint. An equivalence class is necessarily non-empty.

Then we define the quotient set:

$$X/ \equiv := \{cl_{\equiv}(x) \mid x \in X\}$$

And we define the projection function $\pi : X \rightarrow X/ \equiv, \pi(x) := cl_{\equiv}(x)$.

Exercise 5.2. *The quotient set is a set consisting of subsets of the original set, so it's a collection. Also, it is a partition (see the previous example). The quotient set corresponding to the equivalence relation induced by a partition is the original partition; the quotient set of an equivalence relation, viewed as a partition, corresponds to the original equivalence relation.*

Please verify.

Example 5.3. *The quotient set of the integers modulo n is a collection consisting of n elements. The quotient set of the real numbers under the equivalence relation $a \equiv b \Leftrightarrow a - b \in \mathbb{Q}$ is a collection consisting of uncountably many countable sets.*

5.2 Quotient Structures

We often care about some "structures" on sets. Specifically, "structures" are so varied that it's not easy to discuss them generally. Here we demonstrate a few structures and discuss what is required for the corresponding quotient structure.

Example 5.4. *A binary operation can actually be viewed as a 3-ary relation, $R(a, b, c)$ if and only if c is the result of the operation on a and b . An n -ary relation is compatible with an equivalence relation if and only if for $a \equiv a', b \equiv b', \dots$, we have $R(a, b, \dots) \Leftrightarrow R(a', b', \dots)$.*

This already covers the addition, multiplication, and order relation discussed earlier in the context of congruence.

Example 5.5. *Sometimes we care about a function $f : X \rightarrow Y$, consider it as an additional structure on the ordered pair (X, Y) . Suppose we define equivalence relations on the sets X and Y respectively, then we can consider $g : X/\equiv \rightarrow Y/\equiv$ as valid if and only if $\forall x \equiv x', f(x) \equiv f(x')$.*

In summary, if you want to define a function starting from a quotient set, remember to verify $\phi(x) = \phi(x')$ – this is very common in algebra.

5.3 Axiom of Choice

AXIOM 5.1 (Axiom of Choice). *If you have a partition P , i.e., a collection consisting of non-empty, pairwise disjoint sets, then you can take **one** element from each set to form a new set.*

The Axiom of Choice is equivalent to Zorn's Lemma and the Well-Ordering Theorem. We will explain these two theorems later. However, deducing the Well-Ordering Theorem and Zorn's Lemma from the Axiom of Choice requires a certain foundation in modern set theory.

We are introducing the Axiom of Choice here only to answer questions and clarify doubts, and to review partitions and equivalence relations.

Exercise 5.3. *Use the Axiom of Choice to prove: if $g : Y \rightarrow X$ is a surjection, then there exists an injection $f : X \rightarrow Y$. Remember to reduce the discussion to that of set theoretic functions.*

Example 5.6. *In measure theory, we consider the size of subsets of the real numbers. For example, the size of $[a, b]$ is $b - a$, and the size of a singleton set is 0. But not all subsets of real numbers can be reasonably assigned a size.*

Consider the equivalence relation on $[0, 1)$: $a \equiv b \Leftrightarrow a - b \in \mathbb{Q}$. Apply the Axiom of Choice to the quotient set to obtain a set $C \subseteq [0, 1)$. If we hope that cyclic translation preserves the size of the set, and

$$\text{If } S_i \text{ are pairwise disjoint then } \sum_{i=1}^{\infty} |S_i| = \left| \bigcup_{i=1}^{\infty} S_i \right|$$

Then we can deduce a contradiction: $[0, 1)$ can be written as a disjoint union of countably many sets, each obtained by translating C by a rational number, and these sets are all congruent. If C has a size, then the size of $[0, 1)$ is either 0 or ∞ , contradiction.

6 Order and Induction

6.1 Partial Orders, Total Orders, and Basic Examples

We often talk about "order" in daily life, generally referring to a total order. A total order is a binary relation on a set satisfying the following conditions:

Definition 6.1 (Total Order; Partial Order). *A binary relation $\leq$ on a set X is a total order if and only if the following three requirements are met:*

- *Antisymmetry: $\forall x, y \in X, (x \leq y \text{ and } y \leq x) \Leftrightarrow x = y$*

- *Transitivity:* $\forall x, y, z \in X, (x \leq y \text{ and } y \leq z) \Rightarrow x \leq z$
- *Totality:* $\forall x, y \in X, x \leq y \text{ or } y \leq x$

A binary relation satisfying only the first two is called a partial order.

Example 6.1. • The usual order on the real numbers is a total order. A subset of a totally ordered set can be viewed as a totally ordered set – the rational numbers, integers, and natural numbers are all totally ordered sets.

- Consider the set of natural numbers. We say $a \preceq b$ if and only if a divides b , i.e., there exists a natural number n such that $an = b$. $\preceq$ constitutes a partial order on $\mathbb{N}$, which we call the divisibility partial order.
- For a set X , $\subseteq$ constitutes a partial order on $\mathcal{P}(X)$.

Definition 6.2 (Maximal Element, Maximum Element). These two names are hard to remember; unless it's for an exam, I just look them up.

A maximal element is an element in a partially ordered set that has no greater element, while a maximum element is required to be greater than all other elements. Similarly, minimal and minimum elements can be defined.

Example 6.2. In the divisibility partial order on the natural numbers, the maximum element is 0 and the minimum element is 1. If we exclude them and consider the set of natural numbers greater than or equal to 2 with the divisibility partial order, then there is no minimum element, no maximum element, and no maximal element, but minimal elements exist – the minimal elements are exactly prime numbers.

Let's also define some simple small concepts:

Definition 6.3 (Chain). A chain in a partially ordered set is a non-empty totally ordered subset.

Definition 6.4 (Upper Bound). For a subset S of a partially ordered set $(X, \preceq)$, we say x is an upper bound of S if and only if $\forall s \in S, s \preceq x$. An upper bound does not necessarily exist.

Definition 6.5 (Supremum). *For a subset S of a partially ordered set $(X, \preceq)$, we say x is the supremum of S if and only if x is the **least** upper bound of S . Even if an upper bound exists, a supremum does not necessarily exist.*

Definition 6.6 (Lower Bound, Infimum). *Similarly defined.*

In fact, partial orders are everywhere in maths – whenever you define “subsets satisfying certain conditions”, the $\subseteq$ relation appears as a partial order, so they are ubiquitous. So, let us welcome the extremely powerful lemma, Zorn’s Lemma!

6.2 Zorn’s Lemma

Theorem 6.1 (Zorn’s Lemma). *For a non-empty partially ordered set $(X, \preceq)$, if every chain has an upper bound, then this partially ordered set must have a maximal element.*

Clearly, we cannot prove this thing in naive set theory. In general axiomatic set theory systems, it is equivalent to the Axiom of Choice and the Well-Ordering Theorem. To start from the Axiom of Choice, the suggested process is first to prove a set theory theorem about ordinals called Hartogs’s Lemma, then prove the Well-Ordering Theorem, and finally prove Zorn’s Lemma.

However, using Zorn’s Lemma to prove the other two theorems is much simpler. But since we haven’t discussed well-orders yet, let’s try the Axiom of Choice first!

Proposition 6.3. *Zorn’s Lemma can be used to prove the Axiom of Choice.*

Proof. Given a partition P . Consider all sets obtained by “taking **at most one** element from each set in the partition, forming a new set”. The sets obtained in this way form a collection $\mathcal{C}$. Consider $\subseteq$ as the partial order on $\mathcal{C}$.

First, $\mathcal{C}$ is non-empty because $\emptyset \in \mathcal{C}$. Second, all chains in $\mathcal{C}$ have an upper bound: suppose T is a totally ordered subset of $\mathcal{C}$, then $\bigcup T$ can serve as an upper bound for T .

Therefore, Zorn’s Lemma tells us that $\mathcal{C}$ has a maximal element. The maximal element must have selected one element from each set in P , so the Axiom of Choice holds. $\square$

The next time you encounter Zorn's Lemma will probably be in linear algebra, proving that every vector space has a basis – because I've outlined the proof of the Well-Ordering Theorem using the Axiom of Choice, which is coming right up.

6.3 Lots of Mathematical Induction

Example 6.4. *The Peano axiom system uses letters to represent natural numbers variables. Its mathematical symbols are very limited: $0, S, +, \times$. Here, Sn is generally seen as $n + 1$, but in the Peano axioms, we first characterize the behavior of S , then define $+, \times$ using S . Using this symbolism, suppose $\phi(x)$ is a proposition about a natural number x . We can write mathematical induction as:*

$$(\phi(0) \text{ and } (\forall n, \phi(n) \Rightarrow \phi(Sn))) \Rightarrow \forall n, \phi(n)$$

The Peano axiom system includes mathematical induction. We will use mathematical induction and the definition of addition to prove the commutative law of addition. The axioms defining addition in the Peano system are as follows:

- $\forall x, x + 0 = x$
- $\forall x, y, x + Sy = S(x + y)$

Proof. First, prove two lemmas: $\forall x, 0 + x = x$ and $\forall x, y, Sx + y = S(x + y)$. For the former, consider $\phi(x) : \Leftrightarrow 0 + x = x$. First, by the axiom, take $t = 0$, since $t + 0 = 0$, so $0 + 0 = 0$, thus $\phi(0)$ holds. Then assume $\phi(n)$ holds, consider $\phi(Sn)$:

$$0 + Sn = S(0 + n) = Sn$$

Therefore, by mathematical induction, $\forall x, 0 + x = x$.

Next, consider $\psi(y) : \Leftrightarrow \forall x, Sx + y = S(x + y)$. First, $\psi(0)$ is obvious from the axiom. Assume $\psi(n)$ holds, consider $\psi(Sn)$:

$$Sx + Sn = S(Sx + n) = S(S(x + n)) = S(x + Sn)$$

By induction, the lemma holds.

In summary, to prove commutativity $\forall x, y, x + y = y + x$, we just need to perform induction on one variable arbitrarily; here we choose x . First, $\forall y, 0 + y = y = y + 0$. Assume $\forall y, n + y = y + n$ holds, then:

$$Sn + y = S(n + y) = S(y + n) = y + Sn$$

Therefore, by induction, the proof is complete. □

If interested, you can try searching for the Natural Number Game, an interactive web game; it's quite fun, with a first-time playthrough ranging from 3 to 10 hours.

We just demonstrated the most primitive form of mathematical induction. Now let's look at some clever ones.

Proposition 6.5. *Consider the real function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} 0, & x = 0 \\ e^{-1/x^2}, & x \neq 0 \end{cases}$. Prove that f is infinitely differentiable at 0, and all its derivatives at 0 are 0.*

Proof. After trying differentiation, one can summarize that the n -th derivative of f is always a polynomial multiplied by f , so it can be guaranteed that the n -th derivative of f at $x = 0$ is always 0.

Claim: Let P be a polynomial derivative. $g(x) := P(x) * f(x)$, then there exists a polynomial function $Q(x)$ such that $g'(x) = Q(x) * f(x)$. In mathematical analysis, this needs to be discussed separately for $x = 0$ and $x \neq 0$; it is not difficult to prove that this lemma holds.

Therefore, by mathematical induction, the n -th derivative of f is always of the form $P(x) * f(x)$, and its value at 0 is 0. Q.E.D. $\square$

Let's do a puzzling little problem:

Example 6.6. *There is a deck of playing cards, 26 red cards and 26 black cards, no jokers. Now the cards are dealt face down. You can call stop at any time before the last card is dealt. After calling stop, if the next card is red, you win. What is the maximum winning probability your best strategy can achieve?*

Proof. The answer is 50%, strategy has no effect. To discover this, you just need to try what happens when the number of cards is small. To prove this, methods abound; here we explain the simplest and most brute-force method, mathematical induction, which requires no understanding difficulty:

We claim that for this game with n red cards and m black cards, strategy is also useless, and the winning probability is $\frac{n}{n+m}$. First, the base case is when $n + m = 1$, nothing to say, the proposition holds. Then the induction step: assume the proposition holds for $n + m = k$. For given $n = a, m = b, a + b = k + 1$:

Suppose your strategy is to call stop immediately, then the winning probability is directly $\frac{a}{a+b}$. If your strategy is not to call stop

immediately, then after one card is dealt, your strategy becomes a strategy for the $n + m = k$ case, applying the induction hypothesis: If a red card is dealt, the probability of this case is $\frac{a}{a+b}$, and in this case your winning probability is $\frac{a-1}{a+b-1}$; if a black card is dealt, the probability is $\frac{b}{a+b}$, and your winning probability is $\frac{a}{a+b-1}$. Adding them together gives:

$$\frac{a}{a+b} \frac{a-1}{a+b-1} + \frac{b}{a+b} \frac{a}{a+b-1} = \frac{a^2 + ab - a}{(a+b)(a+b-1)} = \frac{a}{a+b}$$

Therefore, strategy is useless in this game, the answer is $1/2$. $\square$

6.4 Well-Ordering and Induction

Definition 6.7 (Transfinite Induction; Well-Founded, Well-Ordering). *Let's rewrite induction. The "next element" is not a widely existing mathematical structure; actually, induction can be reduced to the language of ordering. We can write mathematical induction as:*

$$(\forall n, (\forall t < n, \phi(t)) \Rightarrow \phi(n)) \Rightarrow \forall n, \phi(n)$$

This formula is called transfinite induction. Of course, for the set of natural numbers, this formula is correct. But for the set of real numbers, transfinite induction is not correct:

Let's take $\phi(x) :\Leftrightarrow x \leq 0$. For each $x \in \mathbb{R}$, if $x \leq 0$ then both $(\forall t < x, \phi(t))$ and $\phi(x)$ are true; if $x > 0$, then both small formulas are false, "if (false) then (false)" does not imply the proposition is false. In summary, $(\forall n, (\forall t < n, \phi(t)) \Rightarrow \phi(n))$ indeed holds, but obviously $\forall n, \phi(n)$ does not hold, meaning this induction does not hold for the real numbers.

A partially ordered set for which transfinite induction holds is called "well-founded". A well-founded total order is called a well-order.

Proposition 6.7. *A well-founded partial order is generally defined as a partial order where "every non-empty subset has a minimal element". This definition is completely equivalent to ours.*

Proof. It is not difficult to prove that a well-founded partial order under the general definition satisfies our condition: For a proposition ϕ , consider $S := \{x \mid \phi(x) \text{ is false}\}$. If S has a minimal element, it leads to a contradiction, so S must be empty, hence $\forall x, \phi(x)$.

And it is easy to prove that our partial order satisfies the general condition: For a given subset S , if no minimal element exists, that means $(\forall t < x, t \notin S) \Rightarrow x \notin S$ (otherwise x would be a minimal element). Thus, by induction, S is empty. $\square$

Example 6.8. *The integer order has no least element, so it is clearly not a well-order. The set of natural numbers is a well-ordered set. A finite partially ordered set is always well-founded. $1 - 2^{-n} \mid n \in \mathbb{N} \cup 2 - 2^{-n} \mid n \in \mathbb{N}$, as a subset of the real numbers with the usual order, is well-ordered.*

Well-orders are the shape of logic. If you want to try "take one element, do something, take another, do something, after infinitely many steps take another, and so on ad infinitum", what you need is a well-order. And the Well-Ordering Theorem guarantees that all sets can be well-ordered.

Theorem 6.2. *A partial order is well-founded if and only if there is no infinite descending chain:*

$$a_0 > a_1 > a_2 > \cdots$$

Note the strict inequality $>$, not $\geq$.

Proof. If an infinite descending chain exists, then clearly the set $a_n \mid n \in \mathbb{N}$ has no minimal element, so the partial order is not well-founded. If it is not well-founded, consider a non-empty subset S with no minimal element. Pick $a_0 \in S$, since a_0 is not minimal, we can find $a_1 < a_0$ in S , and so on, obtaining an infinite descending chain. $\square$

Theorem 6.3 (Well-Ordering Theorem). *A set S is said to be well-orderable if and only if there exists a binary relation on S that is a well-order. This theorem claims that all sets are well-orderable.*

The proof details are somewhat intricate; not understanding them is not a big problem.

Proof. We use Zorn's Lemma to prove this theorem... although theoretically it should be the other way around. We cannot directly study the partially ordered set of all well-orderable subsets, as it's not easy to prove that all chains have an upper bound.

Consider the set $\mathcal{P}$ of all ordered pairs $(A, \preceq)$, requiring that $\preceq$ is a well-order on A , and A is a subset of S . Define a partial order on $\mathcal{P}$: $(A, \preceq_1) \leq (B, \preceq_2)$ if and only if:

$$A \subseteq B, \text{ and } (\forall a, a' \in A, a \preceq_1 a' \Leftrightarrow a \preceq_2 a'), \text{ and } (\forall a \in A, \forall b \in B \setminus A, a \preceq b)$$

Intuitively, $\preceq_2$ is an extension of $\preceq_1$ by adding larger elements. Now back to $\mathcal{P}$. First, $(\emptyset, \text{empty relation}) \in \mathcal{P}$ so $\mathcal{P}$ is non-empty. Second, for a chain C , we take the union of all sets as the new subset, and the union of all relations (as subsets of Cartesian products) as the new relation on the new subset. This yields an upper bound for the chain C :

$$C = (A_i, \preceq_i), \preceq_i \subseteq A_i^2.$$

$$A := \bigcup A_i, \preceq := \bigcup \preceq_i$$

Thus, $\mathcal{P}$ has a maximal element. If the corresponding set is not S itself, picking an element not in that subset and adding it as the new greatest element of the well-ordered subset leads to a contradiction. Therefore, S has a corresponding well-order. $\square$

Example 6.9. *Proving the Axiom of Choice using the Well-Ordering Theorem is very simple. Consider a partition P , construct a well-order on $\mathbb{U}P$. Each set in P , as a non-empty subset of $\mathbb{U}P$, has a least element; take these least elements to form a set.*

In fact, the Axiom of Choice has relatively poor universality and average tactical value. The Well-Ordering Theorem is suitable for helping you clarify your thinking, and Zorn's Lemma is suitable for summarizing the problem after clarifying your thinking, capturing the entire problem concisely through a certain degree of abstraction.

6.5 Order Type of Well-Orders

In fact, the "shape" of well-ordered sets is very simple.

Definition 6.8 (Initial Segment). *The initial segment of an element x in a partially ordered set is defined as the subset of elements less than it.*

Definition 6.9 (Order Isomorphism). *Given two partially ordered sets X, Y , an order isomorphism is a bijection $f : X \rightarrow Y$ such that $x_1 \leq x_2$ if and only if $f(x_1) \leq f(x_2)$.*

Given two well-ordered sets, either they are isomorphic, or one is isomorphic to an initial segment of the other.

Don't worry too much about this conclusion; if you study set theory later, you will naturally need to prove it in detail. We won't elaborate further here. If interested, you can try the exercise "Pure Sets and Ordinals" in the exercises.

7 Axiomatization

We have now mastered most of the set theory required for undergraduate studies. Let's complete the final review with axiomatization. We first state the axioms of set theory, then list the basic theory of set theory, and use set theory to define common mathematical structures.

In axiomatic set theory, all mathematical objects are sets, and $\forall, \exists$ also refer to any/there exists a set. Additionally, for convenience, we stipulate the notation $\exists!$ to mean there exists a unique.

AXIOM 7.1 (Axiom Schema of Equality). *This axiom defines the meaning of equality. If $\phi(x)$ is a proposition about x , then we have the axiom:*

$$\forall x \forall y, x = y \Rightarrow (\phi(x) \Leftrightarrow \phi(y))$$

It is called an axiom schema because for different propositions $\phi(x)$, you get a different axiom; this is actually infinitely many axioms, hence called a schema.

AXIOM 7.2 (Axiom of Extensionality). *This axiom defines when sets are equal.*

$$\forall x \forall y, (\forall t, t \in x \Leftrightarrow t \in y) \Rightarrow x = y$$

AXIOM 7.3 (Axiom of Empty Set). *Asserts the existence of the empty set:*

$$\exists e, \forall t, t \notin e$$

Corollary 7.1. *The empty set is unique, provable using the Axiom of Extensionality.*

AXIOM 7.4 (Axiom of Pairing). *Asserts the existence of a, b .*

$$\forall a \forall b, \exists S, (\forall t, t \in S \Leftrightarrow (t = a \text{ or } t = b))$$

AXIOM 7.5 (Axiom of Union). *Asserts the existence of the union of a collection of sets. Combined with the previous axiom, binary union clearly exists, so sets defined by finite listing always exist.*

$$\forall C, \exists U, \forall t, t \in U \Leftrightarrow (\exists X \in C, t \in X)$$

AXIOM 7.6 (Axiom of Power Set). *Asserts that the power set of any set always exists.*

$$\forall X, \exists P, \forall S, (S \subseteq X \Leftrightarrow S \in P)$$

Corollary 7.2. *The above three axioms assert the existence of certain sets. The Axiom of Extensionality implies that such sets are actually unique, so we can use notation to represent these sets: $a, b, \cup C, \mathcal{P}X$.*

AXIOM 7.7 (Axiom of Infinity). *We clearly need to ensure infinite sets exist. In fact, we mentioned earlier how set theory defines natural numbers: $0 := \emptyset$, $S_n := n \cup \{n\}$, each natural number is the set of smaller natural numbers.*

The Axiom of Infinity asserts the existence of the set of natural numbers – technically, the general form of the infinity axiom only guarantees there is a set containing all natural numbers. For convenience, we define a concept:

$$X \text{ is an inductive set} \Leftrightarrow \emptyset \in X \text{ and } \forall x \in X, Sx \in X$$

where Sx denotes $x \cup \{x\}$.

The specific form of the Axiom of Infinity is: There exists an inductive set.

Definition 7.1. *We generally consider the intersection of all inductive sets to be the set of natural numbers. However, "all inductive sets" clearly does not form a set, so we cannot define the intersection as a collection. The good news is that the intersection of inductive sets is inductive, so we can take the collection of inductive subsets of some inductive set w (a subset of $\mathcal{P}w$), and consider the intersection of this collection – selecting a subset requires the following axiom:*

AXIOM 7.8 (Axiom Schema of Separation). *Given a proposition $\phi(x)$ about x and a set X , we can construct the subset of X $x \in X \mid \phi(x)$ holds, i.e., the axiom:*

$$\forall X, \exists Y, \forall t, (t \in Y \Leftrightarrow t \in X \text{ and } \phi(t))$$

This guarantees the existence of intersections and complements.

Corollary 7.3. *The basic properties of intersection, union, and complement are related only to the Axiom of Extensionality; the Axiom of Extensionality directly links the properties of intersection, union, and complement to the properties of logical symbols. Similarly, after we define functions, the various properties of concepts like the pre-image of a function are directly linked, via the set definition, to the properties of logical symbols; only existence needs to be guaranteed by other axioms.*

Proposition 7.4. *Take the set w asserted by the Axiom of Infinity, and construct the following set, calling ω the set of natural numbers:*

$$I := \{S \in \mathcal{P}w \mid S \text{ is inductive}\}$$

$$\omega := \mathbb{N}I := \{t \in \mathbb{N}I \mid \forall X \in I, t \in X\}$$

We claim: ω is a subset of every inductive set. Let T be an inductive set, then $T \cap \omega$ is an inductive set and is a subset of w , so $\omega \subseteq T \cap \omega$, hence $\omega \subseteq T$.

Now it is easy to prove mathematical induction: if a proposition $\phi(n)$ satisfies the conditions for mathematical induction, then by the Axiom Schema of Separation, take $\{n \in \omega \mid \phi(n)\}$ to get an inductive set. Since ω is a subset of this inductive set, $\forall n \in \omega, \phi(n)$.

AXIOM 7.9 (Axiom Schema of Replacement). *Let $\phi(x, y)$ be a proposition about variables x, y . If for $x \in X$ it can be proved that the y satisfying $\phi(x, y)$ exists uniquely, then the axiom guarantees the existence of a set Y consisting of the y corresponding to $x \in X$:*

$$(\forall x \in X, \exists! y, \phi(x, y)) \Rightarrow \exists Y, \forall t, (t \in Y \Leftrightarrow \exists x \in X, \phi(x, t))$$

Proposition 7.5. *This proposition logically should come after defining functions.*

There exists a set larger than $\mathbb{N}, \mathcal{P}\mathbb{N}, \mathcal{P}\mathcal{P}\mathbb{N}, \dots$. Construct as follows: We use sequences to define a recursive proposition (Note: sequences require defining functions, functions require defining Cartesian products and ordered pairs)

$$\phi(n, S) := \exists a^{\text{set theoretic function}}, \text{dom}(a) = n, a(0) = \mathbb{N} \text{ and } \forall i \in n, a(Si) = \mathcal{P}(a(i))$$

By mathematical induction, for any $n \in \omega$, there exists a corresponding unique S . Let $X = \omega$, obtain the corresponding $Y :=$

$\{\mathbb{N}, \mathcal{P}\mathbb{N}, \mathcal{P}\mathcal{P}\mathbb{N}, \dots\}$. Finally, consider $\mathbb{U}Y$ as a set; it is larger than any finite iterated power set of $\mathbb{N}$.

AXIOM 7.10 (Axiom of Choice). *Given a collection of non-empty pairwise disjoint sets, one can choose one element from each set to form a new set.*

$$\forall P, (\emptyset \notin P \text{ and } (\forall x, y \in P, x \cap y \neq \emptyset \Rightarrow x = y)) \Rightarrow (\exists C \subseteq \mathbb{U}P, \forall X \in P, \exists! t \in X \cap C)$$

AXIOM 7.11 (Axiom of Regularity / Foundation). *The Axiom of Regularity may seem unclear in meaning, but it allows set theory to successfully avoid some easily constructed unprovable problems.*

$$\forall X \neq \emptyset, \exists e \in X, e \cap X = \emptyset$$

If we allowed sets like: $A = A, B = B$, then is $A = B$? This is clearly pathological, and such sets have never been of any use, so set theorists choose to directly forbid such sets.

Proposition 7.6.

$$S \notin S$$

Proof. If $S \in S$, consider the set $X := S$; it violates the Axiom of Regularity. $\square$

Proposition 7.7. *(The formal maturity of this theorem requires the use of sequences.) There is no infinite chain:*

$$S_0 \ni S_1 \ni S_2 \ni \dots$$

Proof. If such a chain exists, then $S_n \mid n \in \mathbb{N}$ would also violate the Axiom of Regularity. $\square$

Definition 7.2. • *Ordered pairs can be defined as $(x, y) := x, x, y$.*

We can also express "p = (a, b)", "p is an ordered pair", "a is the first component of p", "b is the second component of p" as set-theoretic statements.

- *Cartesian product*
- *Set-theoretic functions and n-ary relations*
- *Codomain and functions*

- *Injectons, surjections, bijections, and the size of sets.* We should verify that the "size" of sets, after "quotienting out" "same size", satisfies total order conditions (actually it satisfies well-order conditions, but that's more complex). Several theorems we actually skipped; their proofs might be more complex.
- *Disjoint union and Cartesian product of indexed collections*

Theorem 7.1 (Size of Sets). • Define the size relation between sets via injections; below, $|X| \leq |Y|$ means there exists an injection $X \rightarrow Y$.

- *Discuss the relationship with the existence of surjections.*
- *Verify that "size" satisfies $|X| \leq |X|$, and that set size satisfies transitivity (compose injections).*
- *Theorem: If $|X| \leq |Y|$ and $|Y| \leq |X|$, then there exists a bijection between the two sets.*
- *Theorem (not discussed): For any two sets X, Y , either $|X| \leq |Y|$ or $|Y| \leq |X|$.*
- *Theorem (not discussed): Given any collection of sets, there always exists a smallest set (in terms of size). (The sizes of sets are well-ordered.)*

8 Outlook

That's right, all our mathematics can ultimately be translated into the language of set theory. However, as of now, the set theory we have explained still has some shortcomings, such as not proving the Axiom of Choice and some theorems related to cardinality, but this is not a major problem; these things can always be proven, it just requires some small tricks – needing a deeper study of well-orders, exploring what "ordinals" are, and using ordinals to stratify sets. I actually buried a clever little trick in the axiomatization section, which is recursive definition. Recursive definition is not directly allowed by the axioms, but you can use induction to prove that "a function satisfying the recursive requirement exists uniquely". And the operation of recursive definition can be used to define many

things... addition and multiplication on ω , even Turing machines and the "set-theoretic definition of model theory".

Besides, there are things we can study: what exactly is a proposition, what is a proof? We might be able to give a strict, discrete definition, and then discuss set theory within set theory – what kind of operation is this? This is what the course "Mathematical Logic" will explain in detail: What is first-order logic? What is a proof system in first-order logic? How to describe the truth value of propositions when using first-order logic to describe a set with structure?

It is worth noting that propositional logic, first-order logic, and second-order logic are still mostly discussed in first-order logic. Propositional logic is a simple little toy from classical philosophical logic for beginners to play with. First-order logic is the real logic. Second-order logic seems advanced but actually relies on set theory; it essentially uses external set theory to describe a structure on a set, so there's not much to discuss.

Advanced set theory will use model theory to explore various problems, such as independence issues (forcing), properties of large cardinals, the relationship between the constructible universe and countable ordinals, etc.

But that's for later. Generally, it is recommended to read more undergraduate mathematics before playing with mathematical logic. Although mathematical logic does not heavily depend on conclusions from other fields, the motivation is there, and actually the way of thinking in mathematical logic is interconnected with many areas. Even if you are eager to research it, you might as well level up by fighting monsters first.

9 In-Text Exercise Summary and Answers

Definition 9.1. *A set X is said to be smaller than a set Y if and only if there does not exist a surjection $f : X \rightarrow Y$. In this case, Y is said to be larger than X . If X is not larger than Y , and not smaller than Y , then X and Y are said to be the same size.*

Exercise 9.1. *For finite sets, verify that the above definition conforms to convention.*

Answer 9.1. *No answer provided.*

Exercise 9.2. Based on the previously explained relationship between injections and surjections, we can change the definition above from "no injection" to "no surjection in the opposite direction" – though we need to handle the empty set separately. Please rewrite the definition accordingly.

Answer 9.2. A set X is smaller than a set $Y \neq \emptyset$ if and only if there is no surjection from X to Y . Specifically, $\emptyset$ is smaller than all other sets.

Exercise 9.3. We say a function $f : \mathbb{N} \rightarrow \mathbb{N}$ is decreasing if and only if $\forall n \in \mathbb{N}, f(n) \geq f(n+1)$. All decreasing functions form a set S . Prove that S is the same size as $\mathbb{N}$.

Answer 9.3. First, obviously $|\mathbb{N}| \leq |S|$. Next, construct an injection $\Phi : S \rightarrow \mathbb{N}$:

$$\Phi(f) := 2^{f(0)} \prod_{n=1}^{\infty} p_n^{f(n-1)-f(n)}$$

where p_n denotes the n -th prime number, $p_0 = 2, p_1 = 3, p_2 = 5, p_3 = 7, \dots$. Clearly, the infinite product has only finitely many terms not equal to 1 – if there were infinitely many, it would mean the function decreases infinitely many times, starting from $f(0)$, decreasing by 1 each time, which is impossible since the codomain is natural numbers, contradiction.

Clearly Φ is an injection. Q.E.D.

Proposition 9.1. If set X is not larger than Y , and Y is not larger than Z , then X is not larger than Z . In other words, the composition of surjections is a surjection, and the composition of injections is an injection.

Exercise 9.4. Prove the above proposition.

Answer 9.4. The composition $g \circ f$ of injections f, g is an injection: If $(g \circ f)(a) = (g \circ f)(b)$, then $f(a) = f(b)$ so $a = b$, hence $g \circ f$ is an injection.

The composition of surjections is a surjection: To prove $\exists x, g(f(x)) = z$, since g is surjective, find y such that $g(y) = z$; then since f is surjective, find $x \in X$ such that $f(x) = y$, therefore $g(f(x)) = g(y) = z$.

Exercise 9.5. Prove: $(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A))$.

Answer 9.5.

$$x \in (g \circ f)^{-1}(A) \Leftrightarrow g(f(x)) \in A \quad (5)$$

$$\Leftrightarrow f(x) \in g^{-1}(A) \quad (6)$$

$$\Leftrightarrow x \in f^{-1}(g^{-1}(A)) \quad (7)$$

Q.E.D.

Theorem 9.1 (Inverse Functions and Bijections). *$f : X \rightarrow Y$ is a bijection if and only if there exists a function $g : Y \rightarrow X$ such that g can serve as the inverse of f , satisfying: $f \circ g = id_Y$ and $g \circ f = id_X$*

Exercise 9.6. *Prove the above conclusion.*

Answer 9.6. *If $f : X \rightarrow Y$ is a bijection, obviously the inverse function exists... To be precise, expand to the set-theoretic definition of functions. Consider the set-theoretic function corresponding to f , which is a set consisting of ordered pairs (x, y) . Reverse all pairs to get (y, x) ; the resulting set-theoretic function is called $\hat{g}$. $(\hat{g}, X)$ is a bijection $Y \rightarrow X$ and is the inverse of f .*

This exercise is mainly for proving the other direction. If an inverse function exists, then it is a bijection. First, we prove f is injective – if $f(a) = f(b)$, then $g(f(a)) = g(f(b))$, but since $g \circ f = id_X$, we can deduce $a = b$, therefore by definition f is injective.

Next, prove f is surjective. To find x such that $f(x) = y$, just let $x = g(y)$. So f is surjective. In summary, f is a bijection.

Definition 9.2 (Union of a (Indexed) Collection). *For a non-empty (indexed) collection, we define its union:*

$$\cup \mathcal{C} := \{x \mid \exists X \in \mathcal{C}, x \in X\}$$

$$\bigcup_{i \in I} X_i := \{x \mid \exists i \in I, x \in X_i\}$$

Such a union exists uniquely, its existence guaranteed by axioms.

Exercise 9.7. *Verify that this definition is consistent with binary union.*

Answer 9.7. *Need to verify the equality:*

$$\cup \{a, b\} = a \cup b = \bigcup_{i \in \{x, y\}} X_i (X_x = a, X_y = b)$$

Left equality:

$$x \in \mathbb{U}a, b \Leftrightarrow \exists X \in a, b, x \in X \quad (8)$$

$$\Leftrightarrow \exists X, X = a \text{ or } X = b, x \in X \quad (9)$$

$$\Leftrightarrow x \in a \text{ or } x \in b \quad (10)$$

$$\Leftrightarrow x \in a \cup b \quad (11)$$

Right equality similar:

$$x \in \bigcup_{i \in x, y} X_i \Leftrightarrow \exists i \in x, y, x \in X_i \quad (12)$$

$$\Leftrightarrow x \in X_x \text{ or } x \in X_y \quad (13)$$

$$\Leftrightarrow x \in a \text{ or } x \in b \quad (14)$$

$$\Leftrightarrow x \in a \cup b \quad (15)$$

Example 9.2. In this example, (a, b) denotes an open interval of real numbers.

- $\bigcup_{n \in \mathbb{N}} (0, n) = \mathbb{R}^+$
- $\bigcup_{n \in \mathbb{N}} [-1, -1/n] = [-1, 0)$

Exercise 9.8. Carefully verify the above examples, especially the boundaries of the intervals.

Answer 9.8. Clearly for $x \leq 0$, $\forall n \in \mathbb{N}, x \notin (0, n)$, so $x \notin \bigcup$. For $x > 0$, take $n \in \mathbb{N}, n > x$, then $x \in (0, n)$, so $x \in \bigcup$. Therefore $\bigcup = \mathbb{R}^+$.

The second example is also easy to verify, no need to elaborate.

Exercise 9.9. Rewrite the formulas concerning intersections and unions in the properties of pre-images into versions for infinite intersections and unions, and prove them.

Answer 9.9. Here only the union version is written.

$$f^{-1}(\mathbb{U}C) = \mathbb{U}\{f^{-1}(S) \mid S \in C\}$$

$$f^{-1}\left(\bigcup_{i \in I} X_i\right) = \bigcup_{i \in I} f^{-1}(X_i)$$

First equality:

$$x \in f^{-1}(\cup C) \Leftrightarrow f(x) \in \cup C \quad (16)$$

$$\Leftrightarrow \exists X \in C, f(x) \in X \quad (17)$$

$$\Leftrightarrow \exists X \in C, x \in f^{-1}(X) \quad (18)$$

$$\Leftrightarrow x \in \cup f^{-1}(X) \mid X \in C \quad (19)$$

Second equality:

$$x \in f^{-1}\left(\bigcup_{i \in I} X_i\right) \Leftrightarrow f(x) \in \bigcup_{i \in I} X_i \quad (20)$$

$$\Leftrightarrow \exists i \in I, f(x) \in X_i \quad (21)$$

$$\Leftrightarrow \exists i \in I, x \in f^{-1}(X_i) \quad (22)$$

$$\Leftrightarrow x \in \bigcup_{i \in I} f^{-1}(X_i) \quad (23)$$

Example 9.3. • For a set X , we can define two rather meaningless equivalence relations – one says $x \equiv y$ if and only if $x = y$, the other says $x \equiv y$ always holds.

- It is not difficult to verify that the congruence relation on the integers is an equivalence relation.
- We can define an equivalence relation on the real numbers: $a \equiv b$ if and only if $a - b \in \mathbb{Q}$.
- Suppose there is a collection $\mathcal{C}$ where the sets are pairwise disjoint and $\emptyset \notin \mathcal{C}$. Such a collection is called a **partition**. We can define an equivalence relation on the set $\cup \mathcal{C}$: $a \equiv b$ if and only if $\exists X \in \mathcal{C}, a \in X$ and $b \in X$.

Exercise 9.10. Carefully verify the last two examples.

Answer 9.10. Example on real numbers: First, reflexivity $a - a = 0 \in \mathbb{Q}$. Then symmetry: $\mathbb{Q} \ni a - b = -(b - a)$ so $b - a \in \mathbb{Q}$. Finally transitivity: $c - a = (c - b) + (b - a)$, sum of rationals is rational. Partition example: Reflexivity and symmetry are obvious. Transitivity is guaranteed by the pairwise disjointness of sets – $a \equiv b, b \equiv c$ means $a \in X$ and $b \in X$, $b \in Y$ and $c \in Y$, and $b \in X \cap Y$ so $X = Y$, hence $a \in X$ and $c \in X$, so $a \equiv c$.

Exercise 9.11. *The quotient set is a set consisting of subsets of the original set, so it's a collection. Also, it is a partition (see the previous example). The quotient set corresponding to the equivalence relation induced by a partition is the original partition; the quotient set of an equivalence relation, viewed as a partition, corresponds to the original equivalence relation.*

Please verify.

Answer 9.11. *Consider an original partition $\mathcal{P}$, which defines an equivalence relation $\equiv$, then defines a quotient set Q . Need to prove $\mathcal{P} = Q$. We claim that $\mathcal{P}$ consists of the $cl_{\equiv}(x)$, then by the definition of Q , the proof is complete. To prove this, clearly $\forall x \in \mathbb{U}\mathcal{P}, cl_{\equiv}(x) \in \mathcal{P}$, and if $X \in \mathcal{P}$, since $X \neq \emptyset$, take $x \in X$, easily prove $X = cl_{\equiv}(x)$. Q.E.D.*

Consider the reverse direction: If originally there is an equivalence relation $\sim$ on X , defining a quotient set Q , viewed as a partition defining an equivalence relation $\equiv$ on $\mathbb{U}Q$. First, prove separately $X \subseteq \mathbb{U}Q$ and $\mathbb{U}Q \subseteq X$, so $X = \mathbb{U}Q$. Then:

$$x_1 \sim x_2 \Leftrightarrow cl_{\sim}(x_1) = cl_{\sim}(x_2) \in Q \Rightarrow x_1 \equiv x_2$$

$$x_1 \equiv x_2 \Rightarrow \exists X \in Q, x_1 \in X \text{ and } x_2 \in X \quad (24)$$

$$\Rightarrow \exists X \in Q, X = cl_{\sim}(x_1) = cl_{\sim}(x_2) \quad (25)$$

$$\Rightarrow cl_{\sim}(x_1) = cl_{\sim}(x_2) \quad (26)$$

$$\Rightarrow x_1 \sim x_2 \quad (27)$$

Exercise 9.12. *Use the Axiom of Choice to prove: if $g : Y \rightarrow X$ is a surjection, then there exists an injection $f : X \rightarrow Y$. Remember to reduce the discussion to that of set theoretic functions.*

Answer 9.12. *We need to dig down to set-theoretic functions. Let $S_x := \{(g(y), y) \mid g(y) = x\}$ whose existence is ensured by axiom of separation and replacement. Do axiom of choice to the collection $\{S_x \mid x \in X\}$, and get a set-theoretic function, which is injective, and has domain X . Let it's codomain to be Y , and an injection $X \rightarrow Y$ is found.*

10 End-of-Book Exercise Set

The end-of-book exercises systematically consolidate basic discussions on some topics. It is recommended that readers with time

complete them all to solidify their foundation. Please at least complete the contents on "Basic Theory of Functions and Sets" and "Proofs of Countability" in order to test your basic understandings.

10.0.1 Basic Theory of Functions and Sets

Exercise 10.1. (*) In this exercise, we prove some "obvious statements" – translating set language into logical language according to definitions, then transforming it directly based on logical properties, and finally translating it back into set language. In fact, within a strict logical system, we would strictly define a set of fundamental "logical axioms" and perform logical transformations based on them. Please clearly state the reasoning behind every "because and therefore", just like when you first learned geometry in middle school.

- Given $A \subseteq B$ and $B \subseteq C$, prove $A \subseteq C$.
- Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- Prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Given $A, B \subseteq U$, prove $(U \setminus A) \cup (U \setminus B) = U \setminus (A \cap B)$
- Given $A, B \subseteq U$, prove $(U \setminus A) \cap (U \setminus B) = U \setminus (A \cup B)$
- Given $A \subseteq B$, prove $A \times C \subseteq B \times C$
- Given $A \subseteq B$, prove $\mathcal{P}(A) \subseteq \mathcal{P}(B)$

Exercise 10.2. (**) We explore the relationship between whether $f \circ g$ is injective/surjective and the properties of f and g .

- Given that $f \circ g$ is injective, what can we deduce?
- Given that $f \circ g$ is surjective, what can we deduce?

Prove what "can be deduced" and provide counterexamples for what "cannot be deduced".

Exercise 10.3. (***) Now consider a function $f : A \rightarrow B$.

The preimage $f^{-1}(S) := \{a \in A \mid f(a) \in S\}$ can be viewed as a function that takes a subset of B and outputs a subset of A – i.e., $f^{-1} : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$.

We can also define $f^* : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$, $f^*(S) := \{f(s) \mid s \in S\}$.

Verify the following properties:

- f is injective if and only if $f^{-1} \circ f^* = \text{id}_{\mathcal{P}(A)}$
- f is surjective if and only if $f^* \circ f^{-1} = \text{id}_{\mathcal{P}(B)}$
- f is bijective if and only if f^* is the inverse function of f^{-1} .

Exercise 10.4. In this exercise, we practice using inverse functions to prove bijections.

- (*) Prove $|A \times B| = |B \times A|$.
- (*) Given $|A| = |B|, |C| = |D|$, prove $|A \times C| = |B \times D|$.
- (**) Define $Y^X := \{f \mid f : X \rightarrow Y\}$. Given $|A| = |B|, |C| = |D|$, prove $|A^C| = |B^D|$.
- (***) Now consider two indexed families $(X_i)_{i \in I}, (Y_j)_{j \in J}$. Given that $\beta : I \rightarrow J$ is a bijection, and $\forall i \in I, |X_i| = |Y_{\beta(i)}|$, prove $|\bigsqcup_{i \in I} X_i| = |\bigsqcup_{j \in J} Y_j|$, and $|\prod_{i \in I} X_i| = |\prod_{j \in J} Y_j|$.

10.0.2 Countability Proofs and Cardinality Theory

In cardinality theory, we are missing two major theorems:

Exercise 10.5. (***) Given injections $f : X \rightarrow Y$ and $g : Y \rightarrow X$, prove there exists a bijection $h : X \rightarrow Y$.

The proof for this problem is almost pictorial...

Proposition 10.1. Given sets X and Y , prove that either there exists an injection $f : X \rightarrow Y$, or there exists an injection $g : Y \rightarrow X$.

This is best proven using ordinals and the Well-Ordering Theorem; we will cover it in the relevant section.

Let's return to the discussion of countable sets.

Exercise 10.6. (*) Given A, B are both countable sets. Prove $A \times B$ is countable.

Exercise 10.7. (***) Given $\forall i \in \mathbb{N}, A_i$ is countable, and $i \neq j \rightarrow A_i \cap A_j = \emptyset$. Using the Axiom of Choice, prove $\bigcup_{i \in \mathbb{N}} A_i$ is countable, and identify exactly where the Axiom of Choice is used.

Exercise 10.8. (***) Prove that the set of continuous functions from $\mathbb{R} \rightarrow \mathbb{R}$ has a bijection with $\mathbb{R}$. You only need to know that a continuous function is always determined by its values on the rationals, and the set of rationals is countable.

Exercise 10.9. (****) *This problem is not closely related to set theory. However, it's a classic tricky problem, especially when you don't know what it's testing.*

If you know about open sets, a "figure-8" here refers to a shape that divides the plane into three disjoint open sets. If you don't know, it doesn't matter; it just means such a shape, which can be twisted and turned, is in the topological sense. You can use basic topological intuition for this problem without proof.

Prove that the plane cannot contain uncountably many disjoint figure-8s. Below is a hint that reduces the difficulty to two stars. Copy it to a notepad or chat software to view.

(**) 

10.0.3 Pure Sets and Ordinals

In this section, we work within the framework of axiomatic set theory, and all sets considered are pure sets – sets built entirely from other sets, without involving other mathematical concepts. For example, the existence of the empty set does not depend on any other mathematical object, and $\{\emptyset\}$ is also purely constructed from sets. This section will use the concept of "transitive sets" to delve deeper into the structure of well-orders. These exercises are excellent for solidifying logical foundations – at least complete the problems before the five-star ones.

Definition 10.1 (Transitive Set). *A set is transitive if and only if all its elements are also its subsets.*

Exercise 10.10. (**) *Write down a non-empty transitive set, and write down a pure set that is not transitive.*

Exercise 10.11. (*) *We recall that in set theory, natural numbers are represented as pure sets, namely the set of natural numbers less than themselves. That is:*

$$0 = \emptyset, S(n) = n \cup \{n\}$$

Prove that this representation of natural numbers yields transitive sets.

Exercise 10.12. (***) *For a pure set X , find the "smallest" transitive set $Y \supseteq X$. Here "smallest" refers to the $\subseteq$ partial order*

sense, meaning any other set Z satisfying the condition also satisfies $Z \supseteq Y$.

Definition 10.2 (Ordinal). A set is called an "ordinal" if and only if it is a transitive set and all its elements are transitive sets.

Exercise 10.13. (*) Find a transitive set that is not an ordinal.

Exercise 10.14. (*) Prove that the pure set representations of natural numbers mentioned earlier are always ordinals.

Exercise 10.15. (**) Find two infinite ordinals – sets that are infinite.

Exercise 10.16. (*) Prove that the elements of an ordinal are themselves ordinals.

For any natural number n , there is only one ordinal that, as a set, has n elements – namely the set representation of n . We are not in a hurry to prove this proposition; let's look at other things first.

Exercise 10.17. Prove the following properties in order:

- (**) Given a set S of ordinals, $\bigcup S$ is an ordinal.
- (**) Given a non-empty set S of ordinals, $\bigcap S$ is an ordinal.
- (*) Given a set S of ordinals, define the partial order $a \preceq b :\Leftrightarrow a \in b \vee a = b$. Verify it is a partial order. By the Axiom of Regularity, this partial order is well-founded.
- (*****) Prove that the above order is a total order, hence a well-order. Below is a hint breaking this problem into lower-difficulty parts; copy the text to a notepad or chat software to view:
 - Assume a, b are incomparable, i.e., $a \notin b, a \neq b, b \notin a$, denoted $ic(a, b)$.
 - (***) [REDACTED]
 - (***) [REDACTED]
- (*) Based on the previous problem, prove that for ordinals a, b , $a \subsetneq b \Leftrightarrow a \in b$.

In conclusion, $\in$ (which is equivalent to $\subsetneq$) defines a well-order on the ordinals.

We have explored the properties of ordinals. Next, we discuss the role of ordinals. Ordinals are essentially a canonical way to express well-orders: just as we use natural numbers to express the size of finite sets, we use ordinals to express the "shape" of a well-order – two well-orders are isomorphic if and only if their corresponding ordinals are equal.

Definition 10.3 (Order Isomorphism). *Given two partially ordered sets $(X, \leq)$, $(Y, \preceq)$. They are called isomorphic if and only if there exists a bijection $f : X \rightarrow Y$ satisfying:*

$$\forall x_1, x_2 \in X, f(x_1) \preceq f(x_2) \Leftrightarrow x_1 \leq x_2$$

Such a bijection is called an order isomorphism.

Remark 10.2. *Ordinals themselves are well-ordered sets – the elements of an ordinal are ordinals, and $\in$, or equivalently $\subsetneq$, defines a well-order on any set of ordinals.*

Exercise 10.18. *(**) Different ordinals, as well-ordered sets, are non-isomorphic.*

Definition 10.4 (Initial Segment). *We gave this definition before. For a partially ordered set $(X, \leq)$ and $x \in X$, define the initial segment of x :*

$$X^{<x} := \{t \in X \mid t < x\}$$

$X^{<x}$ clearly forms a well-ordered set under $\leq$.

Exercise 10.19. *(***) For any well-ordered set $(X, <)$, there always exists a unique ordinal α isomorphic to it. The idea for this problem is not difficult, but it involves rather intricate details.*

Next, we prove Hartog's Lemma and the Well-Ordering Theorem:

Exercise 10.20. *(Use of Axiom of Choice, Well-Ordering Theorem, Zorn's Lemma is prohibited) (****) Prove Hartog's Lemma: For any set X , there exists an ordinal α such that there is no injection from α into X .*

Exercise 10.21. *(Well-Ordering Theorem) (****) Using the Axiom of Choice and Hartog's Lemma, prove the Well-Ordering Theorem:*

$$\forall X, \exists \preceq \subseteq X^2 \text{ such that } \preceq \text{ is a well-order on } X$$

Exercise 10.22. (**) Prove: For any sets X, Y , either there exists an injection $X \rightarrow Y$, or there exists an injection $Y \rightarrow X$.

Finally, we discuss uncountable ordinals and cardinals, and present another way to understand the Axiom of Regularity.

Definition 10.5 (Cardinal; Countable Ordinal). *An ordinal is called a cardinal if and only if any ordinal less than it is strictly smaller as a set. For example, all natural numbers are cardinals, ω is a cardinal, but $\omega + 1$ is not.*

A countable ordinal generally refers to an ordinal that, as a set, is countably infinite. Natural numbers already have a name, so they are not called countable ordinals. An uncountable ordinal refers to an ordinal that is uncountable as a set.

Exercise 10.23. (***) Prove: The set of natural numbers and all countable ordinals forms an uncountable set, which is an ordinal. This ordinal is denoted ω_1 (standard set theory notation), is the smallest uncountable ordinal, and is also a cardinal.

Remark 10.3. *If you take any sequence of ordinals in ω_1 , it always has a supremum. This means the concept of a sequence is insufficient to exhaust the limit concepts within uncountable ordinals.*

We present an intuitive understanding of the Axiom of Regularity, based on the concept of the Von Neumann hierarchy:

Definition 10.6 (Von Neumann Hierarchy). *Intuitively, we define the Von Neumann hierarchy:*

$$V_\alpha := \bigcup_{\beta < \alpha} \mathcal{P}V_\beta$$

It's easy to see $V_0 = \emptyset$, $V_1 = \{\emptyset\}$, $V_2 = \{\emptyset, \{\emptyset\}\}$, $V_3 = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$, V_4 has 16 elements, V_5 has 65536 elements, V_ω is countable, and sets later are not countable.

To strictly write the proposition $\phi(S, \alpha) : \Leftrightarrow S = V_\alpha$, we can use the technique employed in the proof of the Well-Ordering Theorem.

Definition 10.7 (Rank of a set). *For every set X , there exists an ordinal α such that $X \in V_\alpha$. The smallest ordinal satisfying this condition is called the rank of the set X .*

Exercise 10.24. (**) Using the Axiom of Regularity, prove the above proposition.

10.0.4 Finite Cardinals and Introduction to Combinatorics

In this section, we still adopt the set-theoretic representation of natural numbers, viewing a natural number n as the set of natural numbers less than n , and continue using the notation $Sn = n + 1 = n \cup \{n\}$.

Actually, the introduction to combinatorics part can be viewed separately; it's very simple. Various counting methods in elementary math contests, apart from the Addition and Multiplication Principles, essentially involve constructing bijections and equivalence relations between sets. However, for the sake of theoretical completeness, we start from finite cardinals, fully defining finite sets and natural number operations. If you are not interested in this, you can skip to after the Addition and Multiplication Principles.

Exercise 10.25. (***) *Prove that for natural numbers $n < m$, there is no surjection from n onto m . Similarly, that there is no injection from m into n is omitted here.*

Exercise 10.26. (**) *Let $X \subseteq n$. Prove there exists a unique $m \in Sn$ such that there is a bijection between m and X . In other words, bounded subsets of the set of natural numbers are finite.*

Exercise 10.27. (**) *If there is no surjection from n onto X , then there is an injection from n into X .*

Definition 10.8 (Finite). *The following propositions about a set X are equivalent:*

1. X is finite (defined as any other proposition in this list).
2. There is no surjection from X onto $\mathbb{N}$.
3. There is no injection from $\mathbb{N}$ into X .
4. There exists a (unique) natural number n such that there is a bijection between n and X . In this case, n is called the cardinality of X , denoted $|X|$.
5. For any $t \notin X$ (e.g., in axiomatic set theory, often take $t = X$), there is no bijection from X onto $X \cup \{t\}$.
6. For any $t \notin X$, there is no surjection from X onto $X \cup \{t\}$.

Exercise 10.28. (***) *Prove the equivalence of the above propositions (of course, you don't need to prove the first one... it's the definition). One direction is harder; remember to use the results from the previous two exercises.*

Next, we want to prove the correspondence between finite cardinal arithmetic and natural number arithmetic. Before that, we must define and prove the unique existence of natural number operations:

Definition 10.9. $+$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ refers to the function satisfying the following conditions, which exists uniquely:

$$\forall n, m \in \mathbb{N}, n + 0 = n \quad \text{and} \quad n + Sm = S(n + m)$$

Similarly, multiplication is the function satisfying the following conditions, existing uniquely:

$$\forall n, m \in \mathbb{N}, n * 0 = 0 \quad \text{and} \quad n * Sm = n * m + n$$

Similarly, we define exponentiation, factorial, and prove their unique existence. Note exponentiation also has two axioms, with $0^0 = 1$:

$$a^0 = 1, \quad a^{Sn} = a^n * a$$

Exercise 10.29. (**) *Verify that $+$ exists uniquely.*

Remark 10.4. *Using mathematical induction is sufficient to prove the commutative, associative, and distributive laws of addition and multiplication, and basic properties of exponentiation can also be obtained.*

Remark 10.5. *Many concepts can be defined using addition and multiplication: $a \leq b \Leftrightarrow \exists n, a + n = b$. It's also easy to define divisibility, remainder, congruence, etc.*

Remark 10.6. *Finite sums and finite products can be easily defined.*

Next, we prove the Addition and Multiplication Principles.

Exercise 10.30. (*) *If $|A| = n$ and $t \notin A$, prove $|A \cup \{t\}| = Sn$.*

Exercise 10.31. (**) *Prove the Addition Principle: For a finite family of finite sets $(A_i)_{i=0}^{n-1}$,*

$$(\forall i, j \in n, i \neq j \Rightarrow A_i \cap A_j = \emptyset) \Rightarrow \sum_{i \in n} |A_i| = \left| \bigcup_{i \in n} A_i \right|$$

Corollary 10.7. *The specific formulation of the Multiplication Principle can vary. A common one is: If there are n disjoint sets each with m elements, then their union has nm elements. This is just a special case of the Addition Principle. A concise but less handy form is: $|A \times B| = |A||B|$, which is also easy to prove.*

Exercise 10.32. *(**) For finite sets A, B , define the set:*

$$B^A := \{f \mid f : A \rightarrow B\}$$

Prove $|B^A| = |B|^{|A|}$.

Corollary 10.8. *You can use this notation to reconsider when there is no function from A to B .*

Let's also mention the Inclusion-Exclusion Principle:

Proposition 10.9. *The Inclusion-Exclusion Principle can be written as:*

$$\sum_{J \subseteq n} (-1)^{|J|} \left| \bigcap_{j \in J} A_j \right| = 0$$

Here, specifically, $\bigcap_{j \in \emptyset} A_j := \bigcup_{j \in n} A_j$. This is not standard notation but is useful here.

This involves negative numbers; actually, we can translate it into natural number arithmetic. It doesn't matter.

Exercise 10.33. *(**) Prove the Inclusion-Exclusion Principle for $n = 2$:*

$$|A| + |B| = |A \cup B| + |A \cap B|$$

For larger n , it's clearly an iteration of the binary case.

Next, a commonly used lemma:

Proposition 10.10. *If a finite set X has an equivalence relation $\sim$, then:*

$$|X| = \sum_{c \in X/\sim} |c|$$

If, in particular, each equivalence class has the same size N , then:

$$|X| = |X/\sim| \cdot |N|$$

To facilitate discussion of sets of functions, we introduce another simple lemma:

Exercise 10.34. *(**) Let $\beta : X \rightarrow Y$ be a bijection. Prove that $B_l : X^W \rightarrow Y^W$, $B_l(f : W \rightarrow X) := \beta \circ f$ is a bijection. Similarly, prove $B_r : Y^Z \rightarrow X^Z$, $B_r(g : Z \rightarrow Y) := g \circ \beta$ is also a bijection.*

Based on this, we can easily derive the combination formula from the permutation formula.

Exercise 10.35. *Prove sequentially:*

- *(**) The cardinality of the set of bijections from n to n is $n!$.*
- *(**) For $m < n$, the cardinality of the set of injections from m to n is $\frac{n!}{(n-m)!}$.*
- *(*) For $m < n$, the number of subsets of n with size m is $\frac{n!}{(n-m)!m!}$.*

Exercise 10.36. *(*) Prove the Pigeonhole Principle: Given n mutually disjoint sets $(X_i)_{i=0}^{n-1}$, if $\sum_{i \in n} |X_i| > n * k$, then prove $\exists i \in n, |X_i| > k$.*

Exercise 10.37. *(**) Prove the Stars and Bars method ($m \geq n$):*

$$\exists \beta^{bijection} : \{(a_i^{\in \mathbb{Z}^+})_{i=0}^n \mid \sum_{i \in S_n} a_i = Sm\} \rightarrow \{T \subseteq m \mid |T| = n\}$$

11 End-of-Book Exercises with Answer

11.0.1 Basic Theory of Functions and Sets

Exercise 11.1. *(*) In this exercise, we prove some "obvious statements" – translating set language into logical language according to definitions, then transforming it directly based on logical properties, and finally translating it back into set language. In fact, within a strict logical system, we would strictly define a set of fundamental "logical axioms" and perform logical transformations based on them. Please clearly state the reasoning behind every "because and therefore", just like when you first learned geometry in middle school.*

- *Given $A \subseteq B$ and $B \subseteq C$, prove $A \subseteq C$.*
- *Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$*
- *Prove $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$*

- Given $A, B \subseteq U$, prove $(U \setminus A) \cup (U \setminus B) = U \setminus (A \cap B)$
- Given $A, B \subseteq U$, prove $(U \setminus A) \cap (U \setminus B) = U \setminus (A \cup B)$
- Given $A \subseteq B$, prove $A \times C \subseteq B \times C$
- Given $A \subseteq B$, prove $\mathcal{P}(A) \subseteq \mathcal{P}(B)$

Answer 11.1. • The definition of $X \subseteq Y$ is $\forall x \in X, x \in Y$.
Given $\forall a \in A, a \in B$ and $\forall b \in B, b \in C$. Therefore $\forall a \in A, a \in C$, i.e., $A \subseteq C$.

- First, we have:

$$x \in A \cap (B \cup C) \quad (28)$$

$$\Leftrightarrow x \in A \text{ and } x \in B \cup C \quad (\text{Def of intersection}) \quad (29)$$

$$\Leftrightarrow x \in A \text{ and } (x \in B \text{ or } x \in C) \quad (\text{Def of union}) \quad (30)$$

$$\Leftrightarrow (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C) \quad (31)$$

$$(\text{Properties of AND and OR}) \quad (32)$$

$$\Leftrightarrow x \in A \cap B \text{ or } x \in A \cap C \quad (\text{Def of intersection}) \quad (33)$$

$$\Leftrightarrow x \in (A \cap B) \cup (A \cap C) \quad (\text{Def of union}) \quad (34)$$

By the Axiom of Extensionality, the two sets are equal.

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ is almost identical to the previous exercise.
- Proof:

$$x \in (U \setminus A) \cup (U \setminus B) \quad (35)$$

$$\Leftrightarrow (x \in U \text{ and } x \notin A) \text{ or } (x \in U \text{ and } x \notin B) \quad (36)$$

$$(\text{Def of union and set difference}) \quad (37)$$

$$\Leftrightarrow x \in U \text{ and } (x \notin A \text{ or } x \notin B) \quad (\text{Logical property}) \quad (38)$$

$$\Leftrightarrow x \in U \text{ and } (\neg(x \in A \text{ and } x \in B)) \quad (\text{Logical property}) \quad (39)$$

$$\Leftrightarrow x \in U \text{ and } (\neg(x \in A \cap B)) \quad (\text{Def of intersection}) \quad (40)$$

$$\Leftrightarrow x \in U \text{ and } x \notin A \cap B \quad (41)$$

$$\Leftrightarrow x \in U \setminus (A \cap B) \quad (\text{Def of set difference}) \quad (42)$$

- $(U \setminus A) \cap (U \setminus B) = U \setminus (A \cup B)$ is almost identical to the previous exercise.

- By definition, elements of $A \times C$ always appear in the form $(a \in A, c \in C)$. Because $a \in A \subseteq B$, we have $a \in B$, and because $c \in C$, we have $(a, c) \in B \times C$. Therefore, every element of $A \times C$ is always an element of $B \times C$, hence $A \times C \subseteq B \times C$.
- If $S \in \mathcal{P}(A)$, this means $S \subseteq A$. And because $A \subseteq B$, according to the previously proven result, we have $S \subseteq B$. Therefore, by the definition of $\mathcal{P}(B)$, we have $S \in \mathcal{P}(B)$. So every element of $\mathcal{P}(A)$ is always an element of $\mathcal{P}(B)$, yielding $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Exercise 11.2. *(**) We explore the relationship between whether $f \circ g$ is injective/surjective and the properties of f and g .*

- Given that $f \circ g$ is injective, what can we deduce?
- Given that $f \circ g$ is surjective, what can we deduce?

Prove what "can be deduced" and provide counterexamples for what "cannot be deduced".

Answer 11.2. *Given $f \circ g$ is injective, we can deduce that g is injective. If $g(a) = g(b)$ then $f(g(a)) = f(g(b))$, and since $f \circ g$ is injective, it follows that $a = b$. So $g(a) = g(b) \Rightarrow a = b$, meaning g is injective.*

Clearly, f and g may not be surjective; examples are easy to construct. It's also simple for f not to be injective; we just need the "non-injective" part to lie outside the range of g :

$$g : \mathbb{R}^+ \rightarrow \mathbb{R}, g(x) = x$$

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$$

Similarly, if $f \circ g$ is surjective, we can deduce that f is surjective – Let $g : X \rightarrow Y, f : Y \rightarrow Z$. For $z \in Z$, since $f \circ g$ is surjective, there exists $x \in X$ such that $f(g(x)) = z$. Now take $y = g(x) \in Y$, then $f(y) = z$, so $\forall z \in Z, \exists y \in Y, z = f(y)$.

To make g not surjective is also simple, just make the domain of f unnecessarily large:

$$g : \mathbb{Z} \rightarrow \mathbb{R}, g(x) = x$$

$$f : \mathbb{R} \rightarrow \mathbb{Z}, f(x) = \begin{cases} x, & x \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$

Exercise 11.3. (***) Now consider a function $f : A \rightarrow B$.

The preimage $f^{-1}(S) := \{a \in A \mid f(a) \in S\}$ can be viewed as a function that takes a subset of B and outputs a subset of A – i.e., $f^{-1} : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$.

We can also define $f^* : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$, $f^*(S) := \{f(s) \mid s \in S\}$.

Verify the following properties:

- f is injective if and only if $f^{-1} \circ f^* = \text{id}_{\mathcal{P}(A)}$
- f is surjective if and only if $f^* \circ f^{-1} = \text{id}_{\mathcal{P}(B)}$
- f is bijective if and only if f^* is the inverse function of f^{-1} .

Answer 11.3. Clearly, the third statement is a direct consequence of the first two. We verify the first two separately.

Given f is injective, we want to prove for any $S \subseteq A$ that $f^{-1}(f^*(S)) = S$. The inclusion $S \subseteq f^{-1}(f^*(S))$ can be proven without injectivity – if $s \in S$, then $f(s) \in f^*(S)$, so $s \in f^{-1}(f^*(S))$. Using injectivity, we prove $f^{-1}(f^*(S)) \subseteq S$: if $x \in f^{-1}(f^*(S))$ then $f(x) \in f^*(S)$, meaning there exists $s \in S$ such that $f(x) = f(s)$; by injectivity, $x = s$, so $x \in S$.

Given $f^{-1} \circ f^* = \text{id}_{\mathcal{P}(A)}$, proving $f(x) = f(y) \Rightarrow x = y$ is straightforward: $f^{-1}(f^*(\{x\})) = \{x\}$, but since $f(y) \in f^*(\{x\})$, we have $y \in f^{-1}(f^*(\{x\})) = \{x\}$, hence $y = x$.

Similarly, we always have $f^*(f^{-1}(T)) \subseteq T$: if $b \in f^*(f^{-1}(T))$, then $b = f(a)$ for some $a \in f^{-1}(T)$, which means $f(a) \in T$ – so $b = f(a) \in T$, Q.E.D. If f is surjective, take $b \in T$, then there exists $a \in A$ with $f(a) = b$, so $a \in f^{-1}(T)$ and thus $b \in f^*(f^{-1}(T))$. Therefore, equality holds.

If $f^* \circ f^{-1} = \text{id}_{\mathcal{P}(B)}$, then for $b \in B$, to prove there exists $a \in A$ such that $f(a) = b$, is to prove $f^{-1}(\{b\}) \neq \emptyset$ – assuming it is empty would imply $f^*(\emptyset) = \{b\}$, but clearly $f^*(\emptyset) = \emptyset$, a contradiction.

Exercise 11.4. In this exercise, we practice using inverse functions to prove bijections.

- (*) Prove $|A \times B| = |B \times A|$.
- (*) Given $|A| = |B|, |C| = |D|$, prove $|A \times C| = |B \times D|$.
- (**) Define $Y^X := \{f \mid f : X \rightarrow Y\}$. Given $|A| = |B|, |C| = |D|$, prove $|A^C| = |B^D|$.

- (***) Now consider two indexed families $(X_i)_{i \in I}, (Y_j)_{j \in J}$. Given that $\beta : I \rightarrow J$ is a bijection, and $\forall i \in I, |X_i| = |Y_{\beta(i)}|$, prove $|\bigsqcup_{i \in I} X_i| = |\bigsqcup_{j \in J} Y_j|$, and $|\prod_{i \in I} X_i| = |\prod_{j \in J} Y_j|$.

Answer 11.4. • Define $f : A \times B \rightarrow B \times A, f(a, b) := (b, a)$, and $g : B \times A \rightarrow A \times B, g(b, a) := (a, b)$. These two functions are inverses of each other, hence both are bijections, Q.E.D.

- Let $f : A \rightarrow B, g : C \rightarrow D$ be bijections. Define $h : A \times C \rightarrow B \times D, h(a, c) := (f(a), g(c))$, and $i : B \times D \rightarrow A \times C, i(b, d) := (f^{-1}(b), g^{-1}(d))$. It's easy to verify they are inverses, hence both are bijections, Q.E.D.
- Let $f : A \rightarrow B, g : C \rightarrow D$ be bijections. To define a function, we just need to define the output for each input value. Define $h : A^C \rightarrow B^D$ and $i : B^D \rightarrow A^C$:

$$h[\varphi : C \rightarrow A](d) := f(\varphi(g^{-1}(d)))$$

$$i[\psi : D \rightarrow B](c) := f^{-1}(\psi(g(c)))$$

Verify $h \circ i = id_{B^D}$:

$$(h \circ i)[\psi : D \rightarrow B](d) = h[i[\psi]](d) = f(i[\psi](g^{-1}(d)))$$

$$i[\psi](g^{-1}(d)) = f^{-1}(\psi(g(g^{-1}(d)))) = f^{-1}(\psi(d))$$

$$(h \circ i)[\psi](d) = f(f^{-1}(\psi(d))) = \psi(d)$$

$$(h \circ i)[\psi] = \psi$$

Verification complete. Similarly, $i \circ h = id_{A^C}$.

- Recall the definitions:

$$\prod_{i \in I} X_i := \{f : I \rightarrow \bigcup_{i \in I} X_i \mid \forall i \in I, f(i) \in X_i\}$$

$$\bigsqcup_{i \in I} X_i := \{(i, x) \mid i \in I, x \in X_i\}$$

For each i , fix a bijection $\alpha_i : X_i \rightarrow Y_{\beta(i)}$. Then define the bijection:

$$F : \prod_{i \in I} X_i \rightarrow \prod_{j \in J} Y_j, F(f)(j) := \alpha_{\beta^{-1}(j)}(f(\beta^{-1}(j)))$$

It's clear that $F(f)$ satisfies $\forall j \in J, F(f)(j) \in Y_j$, confirming the codomain is correct. We can write the inverse of F :

$$G : \prod_{j \in J} Y_j \rightarrow \prod_{i \in I} X_i, G(g)(i) := \alpha_i^{-1}(g(\beta(i)))$$

It's easy to verify its codomain is correct. Now verify they are indeed inverses:

$$\begin{aligned} (F \circ G)[g](j) &:= \alpha_{\beta^{-1}(j)}(G[g](\beta^{-1}(j))) \\ G[g](\beta^{-1}(j)) &:= \alpha_{\beta^{-1}(j)}^{-1}(g(\beta(\beta^{-1}(j)))) = \alpha_{\beta^{-1}(j)}^{-1}(g(j)) \\ (F \circ G)[g](j) &= \alpha_{\beta^{-1}(j)}(\alpha_{\beta^{-1}(j)}^{-1}(g(j))) = g(j) \\ (F \circ G)[g] &= g \\ F \circ G &= id_{\prod_{j \in J} Y_j} \end{aligned}$$

Similarly, $G \circ F = id_{\prod_{i \in I} X_i}$.

For the disjoint union, the process is almost identical – write the definition, verify the codomain, verify the inverse relationship. We provide the definitions here:

$$\begin{aligned} F : \bigsqcup_{i \in I} X_i &\rightarrow \bigsqcup_{j \in J} Y_j, F(i, x) := (\beta(i), \alpha_i(x)) \\ G : \bigsqcup_{j \in J} Y_j &\rightarrow \bigsqcup_{i \in I} X_i, G(j, y) := (\beta^{-1}(j), \alpha_{\beta^{-1}(j)}^{-1}(y)) \end{aligned}$$

11.0.2 Countability Proofs and Cardinality Theory

In cardinality theory, we are missing two major theorems:

Exercise 11.5. (***) Given injections $f : X \rightarrow Y$ and $g : Y \rightarrow X$, prove there exists a bijection $h : X \rightarrow Y$.

The proof for this problem is almost pictorial...

Answer 11.5. For $x \in X$, we can draw an arrow to $g(f(x))$. Since $g \circ f$ is injective, each element is pointed to by at most one arrow. What are the possible "chains" in X ? There are three types:

- Cycles. n elements form a cycle connected end to end.

- *Rays.* An element x is the starting point, not pointed to by any other element. Then x points to $g(f(x))$, which points to $g(f(g(f(x))))$, and so on *ad infinitum* – it cannot loop back because each element is pointed to by at most one arrow.
- *Lines.* A chain with no beginning or end.

Clearly, if there are only cycles and lines, then $g \circ f$ is a bijection, hence g is surjective, and g is a bijection, Q.E.D. So we need to handle the case of rays.

For rays, we need to modify g to make it surjective. The problem with rays is that the starting point x might not be in the range of g – of course, it's also possible that $x = g(y)$ but y is not in the range of f ; we don't care about that case. We care about how to correct g to make it surjective when x is not in the range of g .

If the starting point x is not in the range of g , we certainly want to define $g'(f(x)) := x$, but this affects elements later in the ray, so we also need to define $g'(f(g(f(x)))) := g(f(x))$, $g'(f(g(f(g(f(x))))) := g(f(g(f(x))))$, $\dots$. The legitimacy of this operation is guaranteed by the Axiom of Mathematical Induction and the Axiom Schema of Replacement.

To summarize, we define $g'(y) = g(y)$ if and only if $g(y)$ belongs to a cycle, a line, or a ray whose starting point is in the range of g ; if $g(y) = (g \circ f)^n x_0$ and x_0 is not in the range of g , then $n \geq 1$, and we can define $g'(y) := f \circ (g \circ f)^{n-1} x_0$.

Through the above argument, it's not difficult to find that g' defined this way is a bijection. Q.E.D.

Proposition 11.1. *Given sets X and Y , prove that either there exists an injection $f : X \rightarrow Y$, or there exists an injection $g : Y \rightarrow X$.*

This is best proven using ordinals and the Well-Ordering Theorem; we will cover it in the relevant section.

Let's return to the discussion of countable sets.

Exercise 11.6. (*) *Given A, B are both countable sets. Prove $A \times B$ is countable.*

Answer 11.6. *Take bijections $f : A \rightarrow \mathbb{N}$, $g : B \rightarrow \mathbb{N}$. Recall the bijection $\beta : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we constructed earlier in the main text. We get a bijection:*

$$h : A \times B \rightarrow \mathbb{N}, h(a, b) := \beta(f(a), g(b))$$

It's easy to verify this is indeed a bijection.

Exercise 11.7. (***) Given $\forall i \in \mathbb{N}, A_i$ is countable, and $i \neq j \rightarrow A_i \cap A_j = \emptyset$. Using the Axiom of Choice, prove $\bigcup_{i \in \mathbb{N}} A_i$ is countable, and identify exactly where the Axiom of Choice is used.

Answer 11.7. For each i , choose a bijection $\alpha_i : A_i \rightarrow \mathbb{N}$. Recall the bijection $\beta : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, and define the bijection:

$$h : \bigcup_{i \in \mathbb{N}} A_i \rightarrow \mathbb{N}, h(a^{\in A_i}) := \beta(i, \alpha_i(a))$$

Easy to prove it's a bijection.

Essentially, we used the Axiom of Choice in the first step – we know A_i is countable, and for each i , the set of bijections from A_i to $\mathbb{N}$ forms a set. We made a choice from the family $\{B_i := \{f \mid f \text{ bijection} : A_i \rightarrow \mathbb{N}\} \mid i \in \mathbb{N}\}$.

Exercise 11.8. (***) Prove that the set of continuous functions from $\mathbb{R} \rightarrow \mathbb{R}$ has a bijection with $\mathbb{R}$. You only need to know that a continuous function is always determined by its values on the rationals, and the set of rationals is countable.

Answer 11.8. So the set of **continuous functions** from $\mathbb{R} \rightarrow \mathbb{R}$ can be viewed as a subset of the set of **functions** from $\mathbb{Q} \rightarrow \mathbb{R}$, denoted $\mathbb{R}^{\mathbb{Q}}$.

And $\mathbb{R}$ has a bijection with $\mathcal{P}(\mathbb{N})$, $\mathbb{Q}$ has a bijection with $\mathbb{N}$, $\mathbb{N} \times \mathbb{N}$ has a bijection with $\mathbb{N}$:

$$|\mathbb{R}^{\mathbb{Q}}| = |(\mathcal{P}(\mathbb{N}))^{\mathbb{Q}}| = |(2^{\mathbb{N}})^{\mathbb{Q}}| = |2^{\mathbb{N} \times \mathbb{Q}}| = |2^{\mathbb{N} \times \mathbb{N}}| = |2^{\mathbb{N}}| = |\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$$

The "exponentiation properties" here are simple, easily proven lemmas with intuitive bijections and inverse functions.

Exercise 11.9. (****) This problem is not closely related to set theory. However, it's a classic tricky problem, especially when you don't know what it's testing.

If you know about open sets, a "figure-8" here refers to a shape that divides the plane into three disjoint open sets. If you don't know, it doesn't matter; it just means such a shape, which can be twisted and turned, is in the topological sense. You can use basic topological intuition for this problem without proof.

Prove that the plane cannot contain uncountably many disjoint figure-8s. Below is a hint that reduces the difficulty to two stars. Copy it to a notepad or chat software to view.

(**)

Answer 11.9. A rational point refers to a point with both coordinates rational. We say two rational points specify a figure-8 if and only if the two points lie in the two distinct bounded regions (the "holes") of the 8.

Assume there are **disjoint** figure-8s in the plane. Then a pair of rational points can specify at most one figure-8. This means there is a surjection from $\mathbb{Q}^4 = (\mathbb{Q}^2)^2$ to the set of figure-8s. Therefore, the set is countable.

11.0.3 Pure Sets and Ordinals

In this section, we work within the framework of axiomatic set theory, and all sets considered are pure sets – sets built entirely from other sets, without involving other mathematical concepts. For example, the existence of the empty set does not depend on any other mathematical object, and $\{\emptyset\}$ is also purely constructed from sets. This section will use the concept of "transitive sets" to delve deeper into the structure of well-orders. These exercises are excellent for solidifying logical foundations – at least complete the problems before the five-star ones.

Definition 11.1 (Transitive Set). A set is transitive if and only if all its elements are also its subsets.

Exercise 11.10. (**) Write down a non-empty transitive set, and write down a pure set that is not transitive.

Answer 11.10. $\{\emptyset\}$ is transitive. We also have $\{\{\emptyset\}, \emptyset\}$, $\{\{\{\emptyset\}\}, \{\emptyset\}, \emptyset\}$. Clearly, $\{\{\emptyset\}\}$ is not transitive.

Exercise 11.11. (*) We recall that in set theory, natural numbers are represented as pure sets, namely the set of natural numbers less than themselves. That is:

$$0 = \emptyset, S(n) = n \cup \{n\}$$

Prove that this representation of natural numbers yields transitive sets.

Answer 11.11. Use induction. First, the empty set has no elements, so it's trivially transitive. Second, if n is transitive, then $S(n) = n \cup \{n\} \supseteq n$. For $t \in S(n)$, if $t = n$ then $t = n \subseteq S(n)$; if $t \in n$ then because n is transitive, $t \subseteq n$, and since $n \subseteq S(n)$, we have $t \subseteq S(n)$, proving $S(n)$ is also transitive. By mathematical induction, Q.E.D.

Exercise 11.12. (***) For a pure set X , find the "smallest" transitive set $Y \supseteq X$. Here "smallest" refers to the $\subseteq$ partial order sense, meaning any other set Z satisfying the condition also satisfies $Z \supseteq Y$.

Answer 11.12. For a set A , define $F(A) := A \cup (\bigcup A)$. Define the sequence of sets $X_0 = X, X_{S(n)} := F(X_n)$.

$$Y := \bigcup_{n=0}^{\infty} X_n$$

Clearly, Y is transitive – for $y \in Y$, there exists $n \in \mathbb{N}$ such that $y \in X_n$, then $y \subseteq \bigcup X_n \subseteq X_{S(n)} \subseteq Y$.

Suppose there is a transitive set $Z \supseteq X$. To prove $Z \supseteq Y$, by the definition of union, it suffices to prove $\forall n \in \mathbb{N}, X_n \subseteq Z$.

First, $X_0 = X \subseteq Z$. Assume $X_n \subseteq Z$, we want to prove $X_{S(n)} = F(X_n) \subseteq Z$. Clearly, if $x \in F(X_n)$, there are two cases: if $x \in X_n$ then $x \in Z$ directly; if $x \in \bigcup X_n$ then there exists $t \in X_n$ such that $x \in t$, and because $t \in X_n \subseteq Z$ and Z is transitive, $x \in t \subseteq Z$ so $x \in Z$. In conclusion, $X_{S(n)} \subseteq Z$ is proven.

Therefore, for any $n \in \mathbb{N}$, $X_n \subseteq Z$, so $Y \subseteq Z$.

Definition 11.2 (Ordinal). A set is called an "ordinal" if and only if it is a transitive set and all its elements are transitive sets.

Exercise 11.13. (*) Find a transitive set that is not an ordinal.

Answer 11.13. $\{\{\{\emptyset\}\}, \{\emptyset\}, \emptyset\}$

Exercise 11.14. (*) Prove that the pure set representations of natural numbers mentioned earlier are always ordinals.

Answer 11.14. We have already proven that natural numbers are transitive sets. And since the elements of a natural number are also natural numbers, the elements of a natural number are transitive sets. So natural numbers are ordinals.

Exercise 11.15. (**) Find two infinite ordinals – sets that are infinite.

Answer 11.15. The set of natural numbers $\mathbb{N}$ is an ordinal – when we want to emphasize its role as an ordinal, we denote it by ω . For an ordinal α , it's not hard to prove $S(\alpha) := \alpha \cup \{\alpha\}$ is also an ordinal, making $S(\omega)$ an infinite ordinal.

Exercise 11.16. (*) Prove that the elements of an ordinal are themselves ordinals.

Answer 11.16. Given an ordinal X . We want to prove that $x \in X$ is an ordinal. First, the elements of x are elements of X , hence are transitive sets. Second, because x is an element of the ordinal X , x is a transitive set. In conclusion, x is a transitive set whose elements are transitive sets, hence an ordinal.

For any natural number n , there is only one ordinal that, as a set, has n elements – namely the set representation of n . We are not in a hurry to prove this proposition; let's look at other things first.

Exercise 11.17. Prove the following properties in order:

- (**) Given a set S of ordinals, $\bigcup S$ is an ordinal.
- (**) Given a non-empty set S of ordinals, $\bigcap S$ is an ordinal.
- (*) Given a set S of ordinals, define the partial order $a \preceq b :\Leftrightarrow a \in b \vee a = b$. Verify it is a partial order. By the Axiom of Regularity, this partial order is well-founded.
- (*****) Prove that the above order is a total order, hence a well-order. Below is a hint breaking this problem into lower-difficulty parts; copy the text to a notepad or chat software to view:
 - Assume a, b are incomparable, i.e., $a \notin b, a \neq b, b \notin a$, denoted $ic(a, b)$.
 - (***) [REDACTED]
 - (***) [REDACTED]
- (*) Based on the previous problem, prove that for ordinals a, b , $a \subsetneq b \Leftrightarrow a \in b$.

In conclusion, $\in$ (which is equivalent to $\subsetneq$) defines a well-order on the ordinals.

Answer 11.17. • First verify $\bigcup S$ is transitive. For $x \in \bigcup S$, take $X \in S$ such that $x \in X$. Because X is an ordinal, X is transitive, so $x \subseteq X \subseteq \bigcup S$. Therefore $\forall x \in \bigcup S, x \subseteq \bigcup S$, so $\bigcup S$ is transitive.

Next verify the elements of $\bigcup S$ are transitive sets. For $x \in \bigcup S$, take $X \in S$ such that $x \in X$. Because X is an ordinal, x is a transitive set. Therefore, the elements of $\bigcup S$ are transitive sets. *Q.E.D.*

- The elements of $\bigcap S$ are clearly transitive sets. For $x \in \bigcap S$, we have $\forall X \in S, x \in X$, so $\forall X \in S, x \subseteq X$, and it's not hard to prove $x \subseteq \bigcap S$. So $\bigcap S$ is indeed an ordinal.
- Use the properties of transitive sets. Omitted for brevity.
- Consider ordinals a, b . We want $a \notin b, a \neq b, a \not\supseteq b$, and then derive a contradiction. In this case, we want a, b to be "minimal" to derive the contradiction.

For convenience, denote $ic(a, b) :\Leftrightarrow a \notin b, a \neq b, a \not\supseteq b$. First, let a be minimal:

$$S := \{s \in Sa \mid \exists t, ic(s, t)\}, \text{ take } a_0 \in S \text{ with } a_0 \cap S = \emptyset$$

Then let b be minimal:

$$T := \{t \in Sb \mid ic(a_0, t)\}, \text{ take } b_0 \in T \text{ with } b_0 \cap T = \emptyset$$

Now we know $ic(a_0, b_0)$, and $\forall x \in a_0, ic(x, b_0)$ is false, $\forall y \in b_0, ic(a_0, y)$ is false.

Since $b_0 \notin a_0$, we have $\forall x \in a_0, x \neq b_0$ and $x \not\supseteq b_0$ – if $x \supseteq b_0$ then by transitivity $b_0 \in a_0$, contradiction. So $\forall x \in a_0, x \in b_0$, meaning $a_0 \subseteq b_0$. Similarly, we can deduce $b_0 \subseteq a_0$, hence $a_0 = b_0$, contradiction.

- Right-to-left is obvious from $a \notin a$. Left-to-right... because one of $a \in b, a = b, a \supseteq b$ must be true, and it can't be $a = b$ or $a \supseteq b$, so it must be $a \in b$.

We have explored the properties of ordinals. Next, we discuss the role of ordinals. Ordinals are essentially a canonical way to express

well-orders: just as we use natural numbers to express the size of finite sets, we use ordinals to express the "shape" of a well-order – two well-orders are isomorphic if and only if their corresponding ordinals are equal.

Definition 11.3 (Order Isomorphism). *Given two partially ordered sets $(X, \leq)$, $(Y, \preceq)$. They are called isomorphic if and only if there exists a bijection $f : X \rightarrow Y$ satisfying:*

$$\forall x_1, x_2 \in X, f(x_1) \preceq f(x_2) \Leftrightarrow x_1 \leq x_2$$

Such a bijection is called an order isomorphism.

Remark 11.2. *Ordinals themselves are well-ordered sets – the elements of an ordinal are ordinals, and $\in$, or equivalently $\subsetneq$, defines a well-order on any set of ordinals.*

Exercise 11.18. *(**) Different ordinals, as well-ordered sets, are non-isomorphic.*

Answer 11.18. *If $a \neq b$, assume WLOG $a \in b$. Suppose there exists an isomorphism $f : a \rightarrow b$. By induction on the elements of a , it's not difficult to show $\forall x \in a, f(x) = x$. But since $a \notin a$, there is no $x \in a$ such that $f(x) = a \in b$, so f is not surjective, contradiction.*

Definition 11.4 (Initial Segment). *We gave this definition before. For a partially ordered set $(X, \leq)$ and $x \in X$, define the initial segment of x :*

$$X^{<x} := \{t \in X \mid t < x\}$$

$X^{<x}$ clearly forms a well-ordered set under $\leq$.

Exercise 11.19. *(***) For any well-ordered set $(X, <)$, there always exists a unique ordinal α isomorphic to it. The idea for this problem is not difficult, but it involves rather intricate details.*

Answer 11.19. *By induction on $x \in X$, prove that there exists a unique ordinal α_x isomorphic to $X^{<x}$.*

First, we need a small lemma to help complete the isomorphism proof: If $a < b \in X$, and for a there exists a unique α_a isomorphic to $X^{<a}$, and there exists an isomorphism $f_b : X^{<b} \rightarrow \alpha_b$, then $f_b(a) = \alpha_a$. The proof of the lemma is not difficult: since $f_b(a)$,

as an ordinal, is isomorphic to $X^{<a}$ (the isomorphism being the restriction of f_b to $X^{<a}$), uniqueness implies $\alpha_a = f_b(a)$.

Now complete the induction. Assume $\forall a < x$, there exists a unique ordinal α_a isomorphic to $X^{<a}$. We want to construct α_x and the corresponding isomorphism. To construct the ordinal α_x , we use the Axiom Schema of Replacement on $\{a \mid a \in X^{<x}\}$. By the unique existence of each α_a , replace it with $\{\alpha_a \mid a \in X^{<x}\} =: A$, and claim that A is an ordinal and isomorphic to $X^{<x}$.

Proving A is an ordinal is straightforward. Clearly, its elements are transitive sets. Let $\alpha_b \in A$. For $\beta \in \alpha_b$, the function f_b from the lemma ensures $f_b^{-1}(\beta) \in X^{<b} \subseteq X^{<x}$, and $\alpha_{f_b^{-1}(\beta)} = \beta$, so β can be expressed in the form α_t for some $t < x$, hence is an element of A . Therefore, α_b is a subset of A .

How to construct the isomorphism? We reduce the problem to the existence of a set-theoretic function. For the isomorphism $f_b : X^{<b} \rightarrow \alpha_b$, we denote its set-theoretic function as $\hat{f}_b$. We claim that the set $f := \bigcup_{t < x} f_t$ is a set-theoretic function and is an isomorphism $X^{<x} \rightarrow A$. First, the lemma shows f is a function. Then, for $a < b < x$, $f(a) = f_{sb}(a) < f_{sb}(b) = f(b)$, so it's strictly increasing. It's clearly injective and bijective. QED.

The proposition at the beginning of the answer is now proven. Of course, we could perform a similar construction again for the entire set to get the ordinal and isomorphism, but that would be too cumbersome. We define $Y := X \cup \{\infty\}$, where ∞ is any mathematical object not in X . Then forcibly define the well-order on Y to be the well-order on X , with the additional rule that ∞ is the greatest element. Such a well-order clearly exists uniquely.

Therefore, $Y^{<\infty}$ has a unique ordinal isomorphic to it, and $Y^{<\infty}$ is just X . QED.

Next, we prove Hartog's Lemma and the Well-Ordering Theorem:

Exercise 11.20. (Use of Axiom of Choice, Well-Ordering Theorem, Zorn's Lemma is prohibited) (****) Prove Hartog's Lemma: For any set X , there exists an ordinal α such that there is no injection from α into X .

Answer 11.20. The idea of the proof is to let α be the supremum of all order types obtainable from X and its subsets. The detailed process requires careful checking that each step adheres to the axioms.

Consider the set of all well-orderings of subsets of X :

$$\mathcal{S} := \{(S, \preceq) \mid S \subseteq X, \preceq \text{ is a well-order on } S\}$$

This set can be obtained by applying the Axiom Schema of Separation to $\{(S, R) \mid S \subseteq X, R \text{ is a relation on } S\}$, and the latter can be obtained using the Axiom Schema of Replacement and the Axiom of Union:

$\mathcal{R}_S := \{(S, R) \mid R \subseteq S^2\}$ is obtained by applying Replacement to $\mathcal{P}R^2$

$\mathcal{R} := \{\mathcal{R}_S \mid S \subseteq X\}$ is obtained by applying Replacement to $\mathcal{P}X$

$$\mathcal{S} = \bigcup \mathcal{R}$$

Now back to $\mathcal{S}$, each well-ordering $(S, \preceq)$ uniquely corresponds to an ordinal $\alpha_{\preceq}$. Using this, we apply the Axiom Schema of Replacement to $\mathcal{S}$, replacing $(S, \preceq)$ with $\alpha_{\preceq}$, obtaining a set of ordinals Γ . Now claim that Γ is an ordinal and there is no injection from Γ into X . To prove Γ is an ordinal, we only need to prove Γ is transitive, i.e., if $x \in \Gamma$, then for any ordinal $y < x$, $y \in \Gamma$. This just requires considering the well-order $(S, \preceq)$ corresponding to x and taking an initial segment corresponding to y to get $y \in \Gamma$.

If there were an injection $f : \Gamma \rightarrow X$, this injection would define a well-order on the range of f with order type Γ , implying $\Gamma \in \Gamma$, a contradiction.

Therefore, Γ is an ordinal, and there is no injection from Γ into X .

Exercise 11.21. (Well-Ordering Theorem) (****) Using the Axiom of Choice and Hartog's Lemma, prove the Well-Ordering Theorem:

$$\forall X, \exists \preceq \subseteq X^2 \text{ such that } \preceq \text{ is a well-order on } X$$

Answer 11.21. Essentially, what we want to do is take a bag (a set) and put elements of X into it one by one. This is equivalent to the existence of a function – a function $f : \mathcal{P}X \setminus \{X\} \rightarrow \mathcal{P}X$ satisfying the following condition is called a choice function:

$$f(S) \supseteq S, \text{ and there exists a unique } x \in X \text{ such that } x \in f(S) \setminus S$$

That is, $f(S)$ adds one element to S .

The existence of a choice function is guaranteed by the Axiom of Choice – simply choose from the following family of sets to obtain the set-theoretic function for f :

$$\{A_S := \{S \cup \{x\} \mid x \in X \setminus S\} \mid S \in \mathcal{P}X \setminus \{X\}\}$$

The existence of this family is proven by replacing each S in $\mathcal{P}X \setminus \{X\}$ with A_S using Replacement. The unique existence of each A_S is obvious, by applying Replacement to $X \setminus S$.

With the choice function, we can proceed by induction. For convenience, we additionally define $f(X) = X$. Let's first demonstrate the idea, then formalize the details – how to align these seemingly simple and intuitive operations with the axioms. Intuitively, we first use Hartog's Lemma to fix an ordinal Γ for which there is no injection into X . Then for each ordinal $\alpha < \Gamma$, we define a subset $S_\alpha \subseteq X$:

$$S_\alpha := \bigcup_{\beta < \alpha} f(S_\beta)$$

Clearly, $S_0 = \emptyset$, $S_1 = f(S_0)$ has 1 element, S_n has n elements, $S_\omega = \bigcup_{n < \omega} S_{n+1} = \bigcup_{n < \omega} S_n$.

We claim that all S_α exist uniquely. Actually, the definition of S_α is quite vexing and complex; we'll discuss it later. Assuming we have proven this, then everything falls into place: If $\forall \alpha < \Gamma, S_\alpha \neq X$, then $g : \Gamma \rightarrow X, g(\alpha) := f(S_\alpha) \setminus S_\alpha$ is an injection, contradiction. So let β be the least ordinal such that $S_\beta = X$. Then for each $x \in X$, we can find a unique ordinal $\alpha_x < \beta$ such that $x = f(S_{\alpha_x}) \setminus S_{\alpha_x}$, and the order relation among the α_x defines a well-order on X , hence X has a well-order.

Now let's return to strictly defining S_α ... What is S_α ? Or rather, under what circumstances can a set be called S_α ? Actually, instead of saying S_α exists, it's better to say there exists a function $S^\alpha : \alpha \cup \{\alpha\} \rightarrow \mathcal{P}X$, intuitively meaning $S(\beta) = S_\beta$, but essentially:

$$\forall \beta \in \alpha \cup \{\alpha\}, S(\beta) = \bigcup_{\gamma < \beta} f(S(\gamma))$$

Well, then we claim that for each $\alpha < \Gamma$, the function S^α exists uniquely. This isn't difficult; simply use transfinite induction to easily prove the existence of the set-theoretic function.

Exercise 11.22. (**) Prove: For any sets X, Y , either there exists an injection $X \rightarrow Y$, or there exists an injection $Y \rightarrow X$.

Answer 11.22. At this point, everything is very simple. Well-order both sets, with order types α_X, α_Y respectively. If $\alpha_X \subseteq \alpha_Y$, then the inclusion map induces an injection $X \rightarrow Y$, and vice versa. By the total order property of ordinals, QED.

Finally, we discuss uncountable ordinals and cardinals, and present another way to understand the Axiom of Regularity.

Definition 11.5 (Cardinal; Countable Ordinal). *An ordinal is called a cardinal if and only if any ordinal less than it is strictly smaller as a set. For example, all natural numbers are cardinals, ω is a cardinal, but $\omega + 1$ is not.*

A countable ordinal generally refers to an ordinal that, as a set, is countably infinite. Natural numbers already have a name, so they are not called countable ordinals. An uncountable ordinal refers to an ordinal that is uncountable as a set.

Exercise 11.23. (***) *Prove: The set of natural numbers and all countable ordinals forms an uncountable set, which is an ordinal. This ordinal is denoted ω_1 (standard set theory notation), is the smallest uncountable ordinal, and is also a cardinal.*

Answer 11.23. *First, take any uncountable set, e.g., $\mathcal{P}\mathbb{N}$, well-order it, and take the corresponding order type Γ . Clearly Γ is an uncountable ordinal. Take the subset of Γ consisting of natural numbers and countable ordinals, and claim it is the smallest uncountable ordinal ω_1 .*

First, prove it contains all natural numbers and countable ordinals. If α is a natural number or a countable ordinal, clearly α cannot be greater than or equal to Γ . By the total order property of ordinals, $\alpha < \Gamma$, i.e., $\alpha \in \Gamma$. By our construction, $\alpha \in \omega_1$, QED.

Prove it is an ordinal. Consider $\alpha \in \omega_1$. Clearly the elements of α are all natural numbers and countable ordinals. Therefore $\alpha \subseteq \omega_1$. Finally, prove it is the smallest uncountable ordinal. Suppose Θ is an uncountable ordinal. We can similarly construct an $\omega'_1 \subseteq \Theta$, and also prove it is the set of all countable ordinals and natural numbers, so $\omega_1 = \omega'_1$, hence $\omega_1 \subseteq \Theta$, QED.

Remark 11.3. *If you take any sequence of ordinals in ω_1 , it always has a supremum. This means the concept of a sequence is insufficient to exhaust the limit concepts within uncountable ordinals.*

We present an intuitive understanding of the Axiom of Regularity, based on the concept of the Von Neumann hierarchy:

Definition 11.6 (Von Neumann Hierarchy). *Intuitively, we define*

the Von Neumann hierarchy:

$$V_\alpha := \bigcup_{\beta < \alpha} \mathcal{P}V_\beta$$

It's easy to see $V_0 = \emptyset$, $V_1 = \{\emptyset\}$, $V_2 = \{\emptyset, \{\emptyset\}\}$, $V_3 = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$, V_4 has 16 elements, V_5 has 65536 elements, V_ω is countable, and sets later are not countable.

To strictly write the proposition $\phi(S, \alpha) : \Leftrightarrow S = V_\alpha$, we can use the technique employed in the proof of the Well-Ordering Theorem.

Definition 11.7 (Rank of a set). *For every set X , there exists an ordinal α such that $X \in V_\alpha$. The smallest ordinal satisfying this condition is called the rank of the set X .*

Exercise 11.24. *(**) Using the Axiom of Regularity, prove the above proposition.*

Answer 11.24. *If all elements of X have a rank, then by the definition of V_α , it's not hard to prove $\text{rank}(X) \leq S(\bigcup\{\text{rank}(x) \mid x \in X\})$.*

Suppose X has no rank. Then we can find $X_1 \in X$ with no rank, and so on $X_2 \in X_1$ with no rank, ad infinitum, yielding an infinite descending chain:

$$X \ni X_1 \ni X_2 \ni \dots$$

The Axiom of Regularity forbids the existence of such an infinite descending chain, contradiction.

If we want to use the Axiom of Regularity directly without invoking the chain concept, we can consider the smallest transitive set T containing X , then consider $S := \{t \in T \mid t \text{ has no rank}\}$. Take $s \in S$ such that $s \cap S = \emptyset$. s has no rank, so all elements of s have no rank. But $s \cap S = \emptyset$ and all elements of s are in T , meaning $\forall x \in s, x$ has a rank. This forces $s = \emptyset$, but $\emptyset$ has a rank (0), contradiction.

11.0.4 Finite Cardinals and Introduction to Combinatorics

In this section, we still adopt the set-theoretic representation of natural numbers, viewing a natural number n as the set of natural numbers less than n , and continue using the notation $Sn = n + 1 = n \cup \{n\}$.

Actually, the introduction to combinatorics part can be viewed separately; it's very simple. Various counting methods in elementary math contests, apart from the Addition and Multiplication Principles, essentially involve constructing bijections and equivalence relations between sets. However, for the sake of theoretical completeness, we start from finite cardinals, fully defining finite sets and natural number operations. If you are not interested in this, you can skip to after the Addition and Multiplication Principles.

Exercise 11.25. (***) *Prove that for natural numbers $n < m$, there is no surjection from n onto m . Similarly, that there is no injection from m into n is omitted here.*

Answer 11.25. *This proposition strongly depends on the concept of natural numbers, so obviously we must use their essence: mathematical induction.*

First, simplify: we only need to prove there is no surjection from n onto S_n – if there were a surjection $f : n \rightarrow m$, then defining $g(k) = \min(f(k), n)$ would yield a surjection from n onto S_n . So, the non-existence of a surjection from n onto S_n implies the non-existence from n onto m .

Now proceed by induction on n . For the base case $n = 0$, there are no functions from the empty set to a non-empty set, let alone surjections. Assume for $n = k$ that there is no surjection from n onto S_n . We want to prove for $n = S_k$ that there is no surjection from S_k onto SS_k .

Suppose $f : S_k \rightarrow SS_k$ is a surjection. We consider cases. If $f(k) = S_k$, then $f|_k$ itself can be seen as a surjection from k onto S_k , contradiction. If $f(k) \neq S_k$, by surjectivity, take $t \in S_k$ such that $f(t) = S_k$; note $t \neq k$. Now define function g :

$$g(x) := \begin{cases} \min(f(x), k), & x \neq t \\ f(k), & x = t \end{cases}$$

It's not hard to prove $g : k \rightarrow S_k$ is a surjection, contradiction: If we want $g(x) = y$, if $y = k$ take $x = t$; if $y < k$, by surjectivity take x such that $f(x) = y$. Since $f(t) = k > y$, we know $x \neq t$, so $g(x) = f(x) = y$. QED.

Exercise 11.26. (**) *Let $X \subseteq n$. Prove there exists a unique $m \in S_n$ such that there is a bijection between m and X . In other words, bounded subsets of the set of natural numbers are finite.*

Answer 11.26. Uniqueness is obvious from the previous exercise; we need to prove existence. This proposition also holds specifically for natural numbers, so of course we use mathematical induction. For the base case $n = 0 \dots X = \emptyset$ and $n = \emptyset$ certainly have a bijection (the empty function).

If the proposition holds for $n = k$, then for $n = Sk$, consider whether $k \in X$. If $k \notin X$, then $X \subseteq k$ has a bijection with some $m \in Sk \subsetneq Sn$, QED. If $k \in X$, then $X \setminus \{k\} \subseteq k$ has a bijection β with some $m \in Sk$. Define bijection $\beta^* : X \rightarrow Sm$:

$$\beta^*(x) := \begin{cases} \beta(x), & x \neq k \\ m, & x = k \end{cases}$$

QED.

Exercise 11.27. (**) If there is no surjection from n onto X , then there is an injection from n into X .

Answer 11.27. Again, induct on n . $n = 0$ is trivial. Assume for $n = k$ the proposition holds for any X . Now consider $n = Sk$ and a set X .

There is no surjection from Sk onto X , so certainly there is no surjection from k onto X . Thus, there is an injection $f : k \rightarrow X$, and f is not surjective. Suppose $x \in X$ is not in the range of f . Define $g(t) := \begin{cases} f(t), & t \in k \\ x, & t = k \end{cases}$ This function is clearly injective.

QED.

Definition 11.8 (Finite). The following propositions about a set X are equivalent:

1. X is finite (defined as any other proposition in this list).
2. There is no surjection from X onto $\mathbb{N}$.
3. There is no injection from $\mathbb{N}$ into X .
4. There exists a (unique) natural number n such that there is a bijection between n and X . In this case, n is called the cardinality of X , denoted $|X|$.
5. For any $t \notin X$ (e.g., in axiomatic set theory, often take $t = X$), there is no bijection from X onto $X \cup \{t\}$.

6. For any $t \notin X$, there is no surjection from X onto $X \cup \{t\}$.

Exercise 11.28. (***) Prove the equivalence of the above propositions (of course, you don't need to prove the first one... it's the definition). One direction is harder; remember to use the results from the previous two exercises.

Answer 11.28. Since we are discussing mappings, the first thing is to check the empty set, then assume $X \neq \emptyset$. For the empty set, it clearly satisfies all conditions; its cardinality is 0.

Now assume $X \neq \emptyset$. First, prove the equivalence of the first two. If there is a surjection from X onto $\mathbb{N}$, the Axiom of Choice allows creating an injection from $\mathbb{N}$ into X . If there is an injection from X into $\mathbb{N}$, then since $X \neq \emptyset$, there is a surjection from $\mathbb{N}$ onto X . Next consider the next proposition. If there is a bijection $\beta : X \rightarrow n$, and since $n \subseteq \mathbb{N}$, the inclusion map $\iota : n \rightarrow \mathbb{N}$ gives $\iota \circ \beta : X \rightarrow \mathbb{N}$ as an injection.

The difficult part is proving the other direction. Assume X has no bijection with any natural number. We can conclude: X has no injection into any natural number. If X had an injection into n , then the image is a subset of n which has a bijection with some $m \in S_n$, so X would have a bijection with m , contradiction. X has no injection into any natural number implies that no natural number (including 0) has a surjection onto X . Using the previous exercise, we deduce that for any natural number n , there is an injection $f_n : n \rightarrow X$. Furthermore, examining the construction in our proof, we can happily find that for $n < m$, f_n is the restriction of f_m , or as set-theoretic functions, $f_m \supseteq f_n$. Then we can see that $\bigcup_{n \in \mathbb{N}} f_n$ is an injection from $\mathbb{N}$ into X .

Now for the last two. If a bijection exists from X to $X \cup \{t\}$, then obviously a surjection exists. If there is a surjection $f : X \rightarrow X \cup \{t\}$, let $t_0 := t$, and for $t_n \in X \cup \{t\}$, take $t_{S_n} \in X$ such that $f(t_{S_n}) = t_n$. It's not hard to prove that for $a < b \in \mathbb{N}$, $t_a \neq t_b$ – just induct on b . Therefore, $i : \mathbb{N} \rightarrow X, i(n) = t_{S_n}$ is an injection.

If there is an injection $\iota : \mathbb{N} \rightarrow X$, then we can simply define a bijection $\beta : X \rightarrow X \cup \{t\}$:

$$\beta(x) := \begin{cases} t, & x = \iota(0) \\ \iota(n), & x = \iota(Sn) \\ x, & x \notin \text{range of } \iota \end{cases}$$

QED.

Next, we want to prove the correspondence between finite cardinal arithmetic and natural number arithmetic. Before that, we must define and prove the unique existence of natural number operations:

Definition 11.9. $+$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ refers to the function satisfying the following conditions, which exists uniquely:

$$\forall n, m \in \mathbb{N}, n + 0 = n \quad \text{and} \quad n + Sm = S(n + m)$$

Similarly, multiplication is the function satisfying the following conditions, existing uniquely:

$$\forall n, m \in \mathbb{N}, n * 0 = 0 \quad \text{and} \quad n * Sm = n * m + n$$

Similarly, we define exponentiation, factorial, and prove their unique existence. Note exponentiation also has two axioms, with $0^0 = 1$:

$$a^0 = 1, \quad a^{Sn} = a^n * a$$

Exercise 11.29. (**) Verify that $+$ exists uniquely.

Answer 11.29. We follow the construction of set-theoretic functions to prove this. Intuitively, we want to prove the existence of the following functions:

$$\forall n, \exists +_n : \mathbb{N} \rightarrow \mathbb{N}, \quad +_n(x) = "x + n"$$

The quotes indicate this is not a strict definition but provides intuitive understanding. Strictly, we require:

$$+_n(0) = n \quad \text{and} \quad +_n(Sm) = S(+_n(m))$$

Clearly, $+_0$ is $\text{id}_{\mathbb{N}}$, which exists and is unique. And if $+_n$ exists and is unique, then by the Axiom Schema of Replacement, $+_{Sn}$ also exists and is clearly unique.

Finally, we can perform some messy operations using the Axiom Schema of Replacement to obtain the existence of $+$. The uniqueness of $+$ can also be proven by induction on n , proving the following proposition:

If f satisfies the conditions, then $g(x) := f(x, n)$ must satisfy $g = +_n$

Remark 11.4. *Using mathematical induction is sufficient to prove the commutative, associative, and distributive laws of addition and multiplication, and basic properties of exponentiation can also be obtained.*

Remark 11.5. *Many concepts can be defined using addition and multiplication: $a \leq b :\Leftrightarrow \exists n, a + n = b$. It's also easy to define divisibility, remainder, congruence, etc.*

Remark 11.6. *Finite sums and finite products can be easily defined.*

Next, we prove the Addition and Multiplication Principles.

Exercise 11.30. *(*) If $|A| = n$ and $t \notin A$, prove $|A \cup \{t\}| = Sn$.*

Answer 11.30. *Take a bijection $\beta : A \rightarrow n$, and construct a bijection $f : A \cup \{t\} \rightarrow Sn$:*

$$f(a) := \begin{cases} \beta(a), & a \in A \\ n, & a = t \end{cases}$$

Exercise 11.31. *(**) Prove the Addition Principle: For a finite family of finite sets $(A_i)_{i=0}^{n-1}$,*

$$(\forall i, j \in n, i \neq j \Rightarrow A_i \cap A_j = \emptyset) \Rightarrow \sum_{i \in n} |A_i| = \left| \bigcup_{i \in n} A_i \right|$$

Answer 11.31. *Assume it holds for $n = k$. For $n = Sk$, define $B = \bigcup_{i \in k} A_i$ and $A = A_k$. This reduces to proving:*

$$A \cap B = \emptyset \Rightarrow |A| + |B| = |A \cup B|$$

That is, the binary case of the Addition Principle. The essence of our discussion is to correlate cardinal addition (i.e., disjoint union) with natural number addition, which inevitably requires using the essential property of natural numbers – mathematical induction.

Induct on $|B|$. If $|B| = 0$, then $A = A \cup B$, QED. If the proposition holds for $|B| = k$, then for $|B| = Sk$, take a bijection $\beta : B \rightarrow Sk$. Let $B' := B \setminus \beta^{-1}(k)$. It's easy to see $|B| = S|B'|$, and $|A \cup B| = S|A \cup B'|$. Also, we have $|A \cup B'| = |A| + |B'|$. Then:

$$|A| + |B| = |A| + S|B'| = S(|A| + |B'|) = S(|A \cup B'|) = |A \cup B|$$

QED.

Corollary 11.7. *The specific formulation of the Multiplication Principle can vary. A common one is: If there are n disjoint sets each with m elements, then their union has nm elements. This is just a special case of the Addition Principle. A concise but less handy form is: $|A \times B| = |A||B|$, which is also easy to prove.*

Exercise 11.32. *(**) For finite sets A, B , define the set:*

$$B^A := \{f \mid f : A \rightarrow B\}$$

Prove $|B^A| = |B|^{|A|}$.

Answer 11.32. *By induction on $|A|$, applying the Multiplication Principle and the definition of exponentiation.*

Corollary 11.8. *You can use this notation to reconsider when there is no function from A to B .*

Let's also mention the Inclusion-Exclusion Principle:

Proposition 11.9. *The Inclusion-Exclusion Principle can be written as:*

$$\sum_{J \subseteq n} (-1)^{|J|} \left| \bigcap_{j \in J} A_j \right| = 0$$

Here, specifically, $\bigcap_{j \in \emptyset} A_j := \bigcup_{j \in n} A_j$. This is not standard notation but is useful here.

This involves negative numbers; actually, we can translate it into natural number arithmetic. It doesn't matter.

Exercise 11.33. *(**) Prove the Inclusion-Exclusion Principle for $n = 2$:*

$$|A| + |B| = |A \cup B| + |A \cap B|$$

For larger n , it's clearly an iteration of the binary case.

Answer 11.33. *I choose to induct on $|A \cap B|$. If it's empty, it's the Addition Principle. The induction step involves taking an element from B that is in $A \cap B$ to get B' , then adding it back:*

$$|A| + |B| = |A| + S|B'| = S(|A| + |B'|) = S(|A \cup B'| + |A \cap B'|) \quad (43)$$

$$= |A \cup B'| + S|A \cap B'| = |A \cup B| + |A \cap B| \quad (44)$$

QED.

Next, a commonly used lemma:

Proposition 11.10. *If a finite set X has an equivalence relation $\sim$, then:*

$$|X| = \sum_{c \in X/\sim} |c|$$

If, in particular, each equivalence class has the same size N , then:

$$|X| = |X/\sim| \cdot |N|$$

To facilitate discussion of sets of functions, we introduce another simple lemma:

Exercise 11.34. *(**) Let $\beta : X \rightarrow Y$ be a bijection. Prove that $B_l : X^W \rightarrow Y^W$, $B_l(f : W \rightarrow X) := \beta \circ f$ is a bijection. Similarly, prove $B_r : Y^Z \rightarrow X^Z$, $B_r(g : Z \rightarrow Y) := g \circ \beta$ is also a bijection.*

Answer 11.34. *The expression for the bijection is very simple and clear; we construct the inverse function. First, let β^{-1} be the inverse of β . Then $I_l : Y^W \rightarrow X^W$, $I_l(f : W \rightarrow Y) := \beta^{-1} \circ f$ is clearly the inverse of B_l :*

$$I_l(B_l(f)) = \beta^{-1} \circ \beta \circ f = id \circ f = f$$

$$B_l(I_l(f)) = \beta \circ \beta^{-1} \circ f = id \circ f = f$$

So B_l is a bijection. Similarly, B_r is a bijection.

Based on this, we can easily derive the combination formula from the permutation formula.

Exercise 11.35. *Prove sequentially:*

- *(**) The cardinality of the set of bijections from n to n is $n!$.*
- *(**) For $m < n$, the cardinality of the set of injections from m to n is $\frac{n!}{(n-m)!}$.*
- *(*) For $m < n$, the number of subsets of n with size m is $\frac{n!}{(n-m)!m!}$.*

Answer 11.35. • *Induct on n . Assume for $n = k$ there are $k!$ bijections. For $n = Sk$, consider that a bijection f has n choices for $f(k)$. For a given $f(k)$, the remaining k elements*

form a bijection $\hat{f} : k \rightarrow n \setminus \{f(k)\}$. The set $n \setminus \{f(k)\}$ has a bijection β with k , giving a bijection between $\{\hat{f} \mid f(k) = a\}$ and $\{\text{bijections } g \mid g : k \rightarrow k\}$. Therefore, for a given $f(k)$, there are exactly $k!$ possible f . So total is $n * k!$ bijections. QED.

- We can turn a bijection $n \rightarrow n$ into an injection $m \rightarrow n$ by ignoring $f(m), f(Sm), \dots, f(n-1)$. It's not hard to see that for a given injection, the set of bijections corresponding to it is in bijection with the set of bijections $(n-m) \rightarrow (n-m)$:

$$|\{f|_m = g \mid f \text{ is a bijection } : n \rightarrow n\}| = |\{h \mid h \text{ is a bijection } : n-m \rightarrow n-m\}|$$

Writing out this bijection and its inverse proves the bijection exists, so the equation holds.

Therefore, each injection $m \rightarrow n$ corresponds to $(n-m)!$ bijections $n \rightarrow n$. Applying the Addition Principle in reverse gives:

$$ANS \cdot (n-m)! = n!$$

Therefore:

$$ANS = \frac{n!}{(n-m)!}$$

- Then, by viewing an injection f as the subset $\{f(x) \mid x \in m\}$, we get a new equation. The situation is similar and won't be elaborated.

Exercise 11.36. (*) Prove the Pigeonhole Principle: Given n mutually disjoint sets $(X_i)_{i=0}^{n-1}$, if $\sum_{i \in n} |X_i| > n * k$, then prove $\exists i \in n, |X_i| > k$.

Answer 11.36. This actually has little to do with sets; it's purely about natural number addition. If $\forall i \in n, |X_i| \leq k$, then $\sum_{i \in n} |X_i| \leq \sum_{i \in n} k = n * k$, contradiction.

Exercise 11.37. (**) Prove the Stars and Bars method ($m \geq n$):

$$\exists \beta^{\text{bijection}} : \{(a_i^{\in \mathbb{Z}^+})_{i=0}^n \mid \sum_{i \in S_n} a_i = Sm\} \rightarrow \{T \subseteq m \mid |T| = n\}$$

Answer 11.37. As taught in elementary math contests, write out β and β^{-1} , verify their codomains, and verify they are inverses:

$$\beta((a_i)) := \left\{ \sum_{i \in Sk} a_i \mid k \in n \right\}$$

Verification: Since $k \in n$ and $a_n > 0$, $\sum_{i \in S_k} a_i < \sum_{i \in S_n} a_i = m$, so $\sum_{i \in k} a_i \in m$, therefore $\beta((a_i)) \subseteq m$. Also need to verify $|\beta((a_i))| = n$:

If $t > 0$ and $\sum_{i \in S_k} a_i = \sum_{i \in S(k+t)} a_i$, this would imply $\forall i \in \{Sk, SSk, \dots k+t\}, a_i = 0$, but $a_{Sk} = 0$ already gives a contradiction. So there's an obvious bijection from n to $\beta((a_i))$: $k \mapsto \sum_{i \in S_k} a_i$.

After verifying the codomain, consider the inverse function:

$$(\beta^{-1}(T))_i := (\text{the } (Si)\text{th smallest element in } T)$$

$$-(\text{the } i\text{th smallest element in } T)$$

Specifically, define the 0th smallest element as -1 , and the (Sn) th smallest element as Sm . Writing this out strictly with symbols would be a bit lengthy, but it's just a small lemma, so won't elaborate. It's not hard to see that all numbers in the sequence are positive integers and sum to $Sm - 0 = Sm$. Codomain verification complete.

Verifying that they are inverse functions is straightforward.